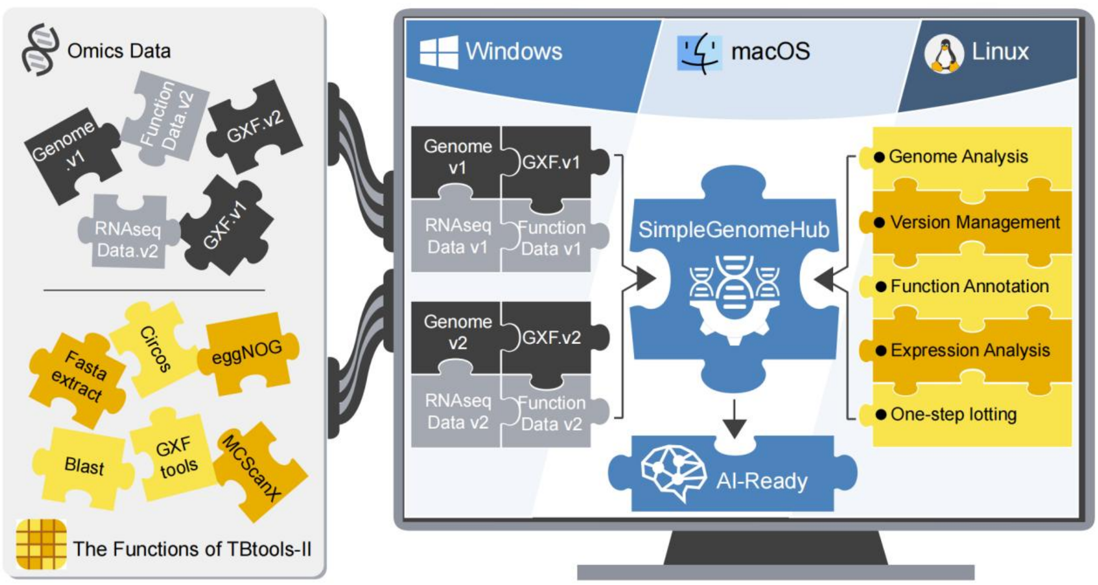
SimpleGenomeHub | 1. Introduction

Simple Genome Hub is a cross-platform tool for the local management and analysis of genomic data. It is designed to help researchers move away from cumbersome manual data-management workflows and to enable easy use without requiring extensive prerequisite knowledge. The software supports standardized management of omics data, efficient mining of biological information, and long-term preservation and retrospective traceability of analytical data.。

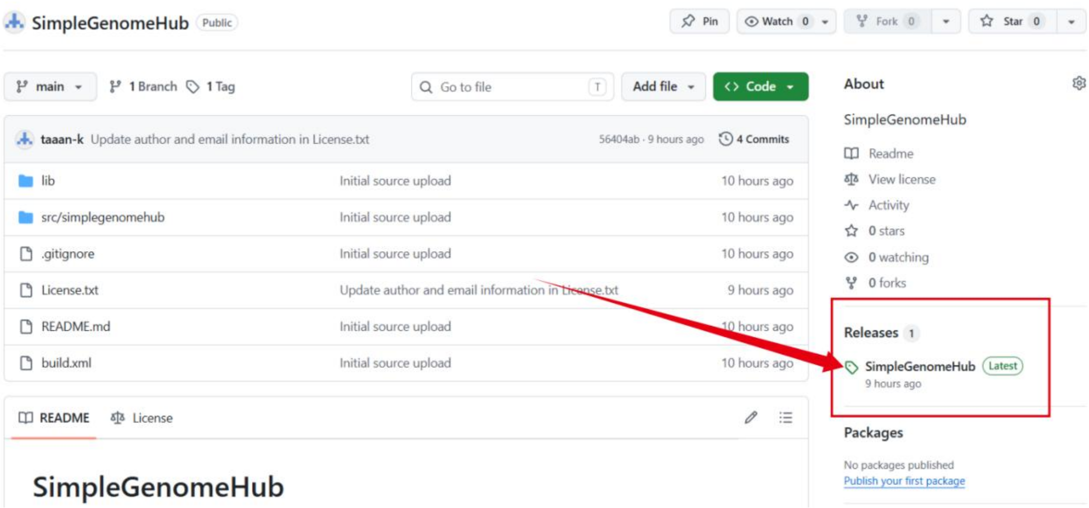
Design Philosophy

- A data-unit management model is adopted in which "**one genome assembly corresponds to one set of annotation information**", thereby standardizing data usage.
- A stable local data directory tree is defined. All data are organized under the main database path as the root directory. After import, data are programmatically archived and can be exported, reducing the uncertainty and burden associated with manual judgment and selection.
- On this basis, a data-centric framework is established by integrating functional modules from **TBtools-II**, forming a convenient, cross-platform data exploration framework in which local data and analytical functions can be directly integrated and invoked.



Download

- The core source code of SimpleGenomeHub has been released as open source in a GitHub repository. The compiled, ready-to-use software can be downloaded from the **Releases** section on the right-hand side of the following page: https://github.com/taaan-k/SimpleGenomeHub/.
- At present, the software needs to be launched by clicking **run.bat** from a directory whose full path contains only English characters. As the current version is still at an early stage, there may be potential bugs or user-unfriendly design issues. We appreciate your understanding, and these issues will be gradually addressed in future updates.



The icons used in the example workflow.

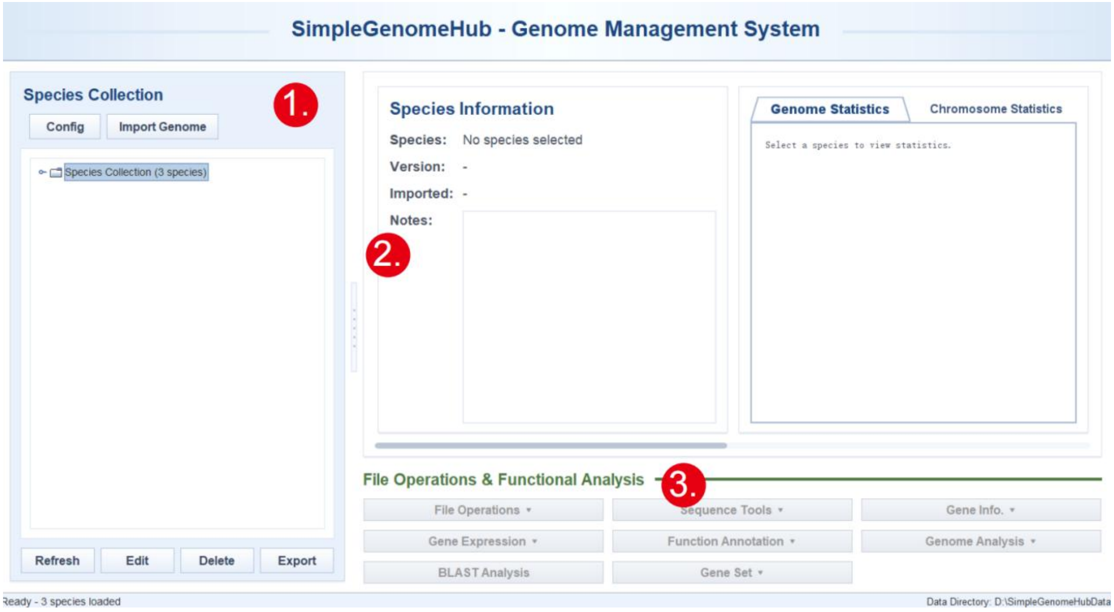
Simple Genome Hub | 2. Interface Design

The main interface consists of three primary panels:

1. Database Management Panel;

2. Basic Information Panel for Data Units;

3. File Operations and Functional Modules Panel.



I. Database Management Panel

This panel mainly includes global settings, data import, data export, data deletion, basic information editing, and an interactive directory tree.

II. Basic Information Panel for Data Units

This panel mainly displays the species information of the selected data unit, genome version and basic information, annotation information, chromosome information, and gene/region sets of interest.

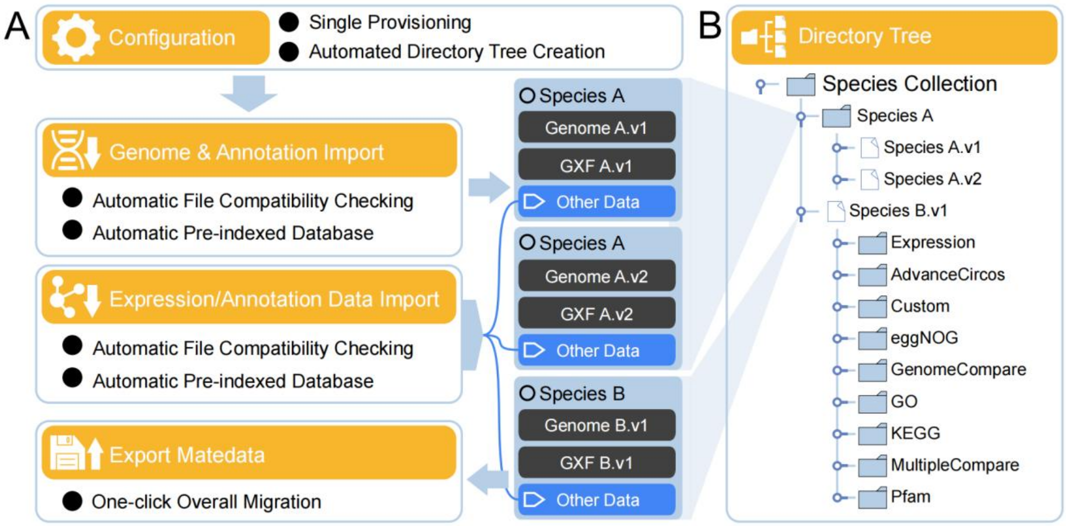
III. File Operations and Functional Modules Panel

This panel contains all available operations and functions, which are organized under eight main buttons.

Simple Genome Hub | 3. Data Management and Gene Sets

A largely automated genome management approach:

- One-time global configuration. The storage path only needs to be configured during the first use. Please ensure that sufficient storage space is reserved, generally at least 20 GB. All genome-related files, including sequences, annotations, expression data, and synteny information, are automatically organized according to the standard data unit of “one genome assembly with one corresponding set of annotations.”
- One-click packaging & migration. During packaging, the software automatically identifies the assigned paths and ensures that all relevant files are included.
- Intuitive data management through a directory tree. The directory tree provides a clear and user-friendly interface for data management and interactive operations.

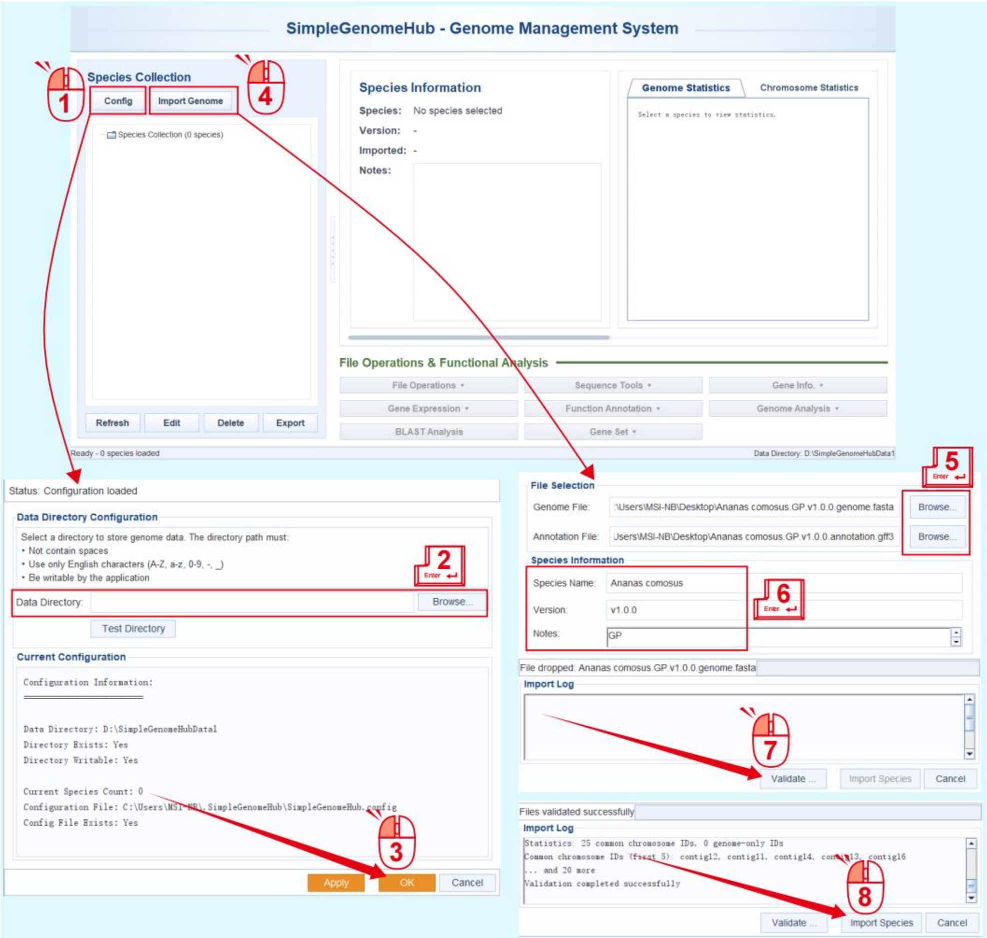


I. Global Configuration and Genome Import

1. When using the software for the first time, click **Config** button to perform global configuration.

2. Configure the storage path and ensure that at least 20 GB of free space is available.

3. Click **OK** to complete the configuration.



After this initial setup, the Config step is no longer required when importing new genomes. New genomes can be imported directly by following the workflow below:

4. Click the **Import Genome** button.

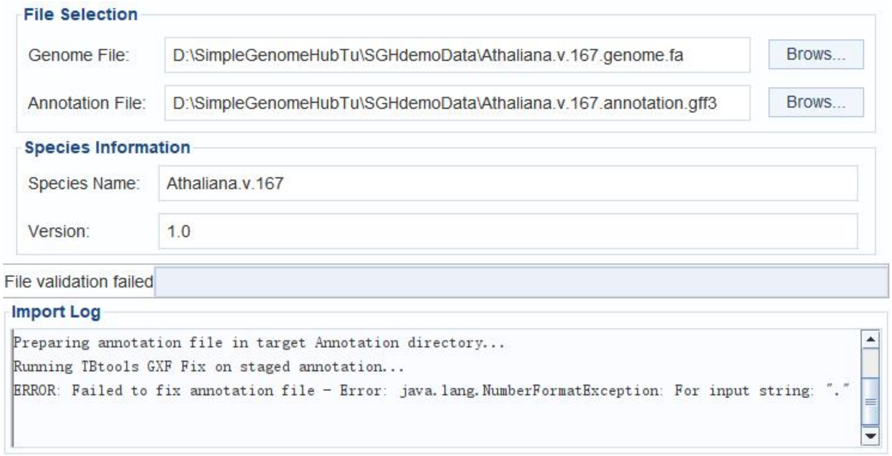
5. Drag and drop the genome sequence file and annotation file to be imported.

6. Edit the genome version, notes, and other related information.

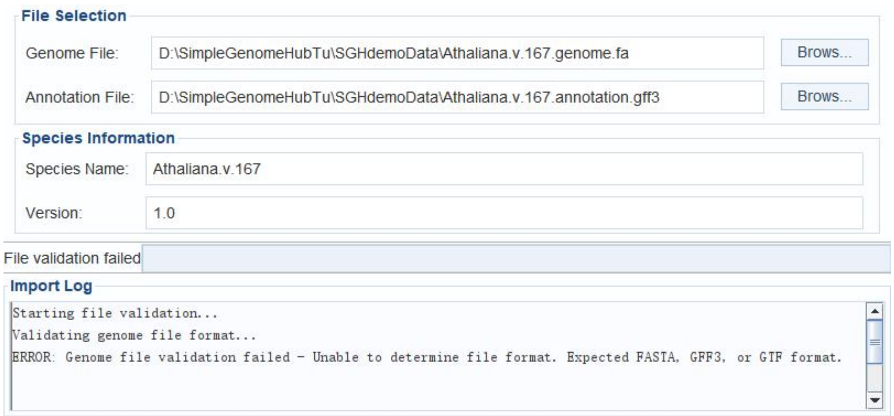
7. Click **Validate...** button to verify whether the genome sequence file and annotation file are matched. The annotation file will also be automatically corrected using the GXF FIX function from TBtools-II. If any problem is detected, a pop-up message will be displayed.

8. After validation is passed, click the **Import Species** button to complete the import.

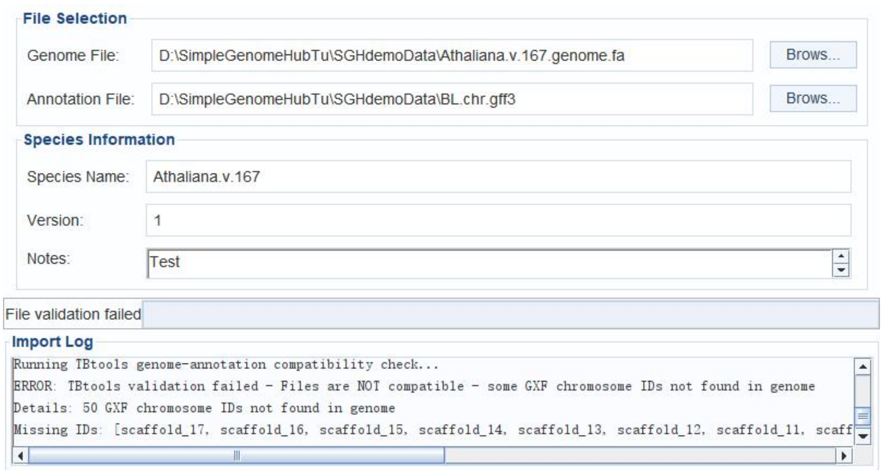
- During validation, SGH checks the format of the annotation file and reports any format errors in the log:



- It also checks the format of the genome file and reports any errors in the log:

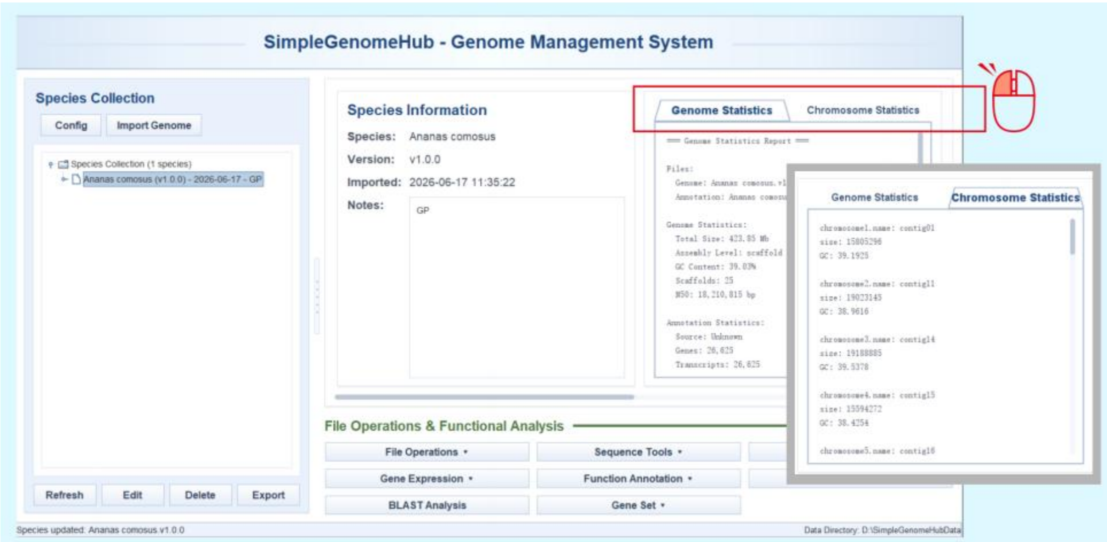


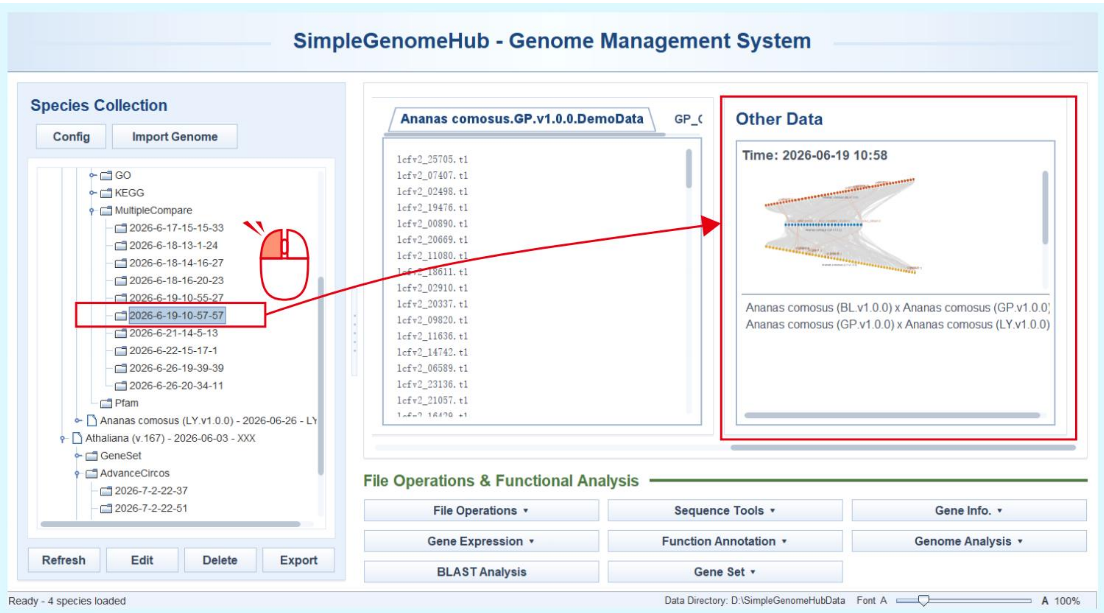
- In addition, SGH checks whether chromosome names match between the genome and annotation files and whether annotated gene positions exceed chromosome lengths. Any detected issues are reported in the log:

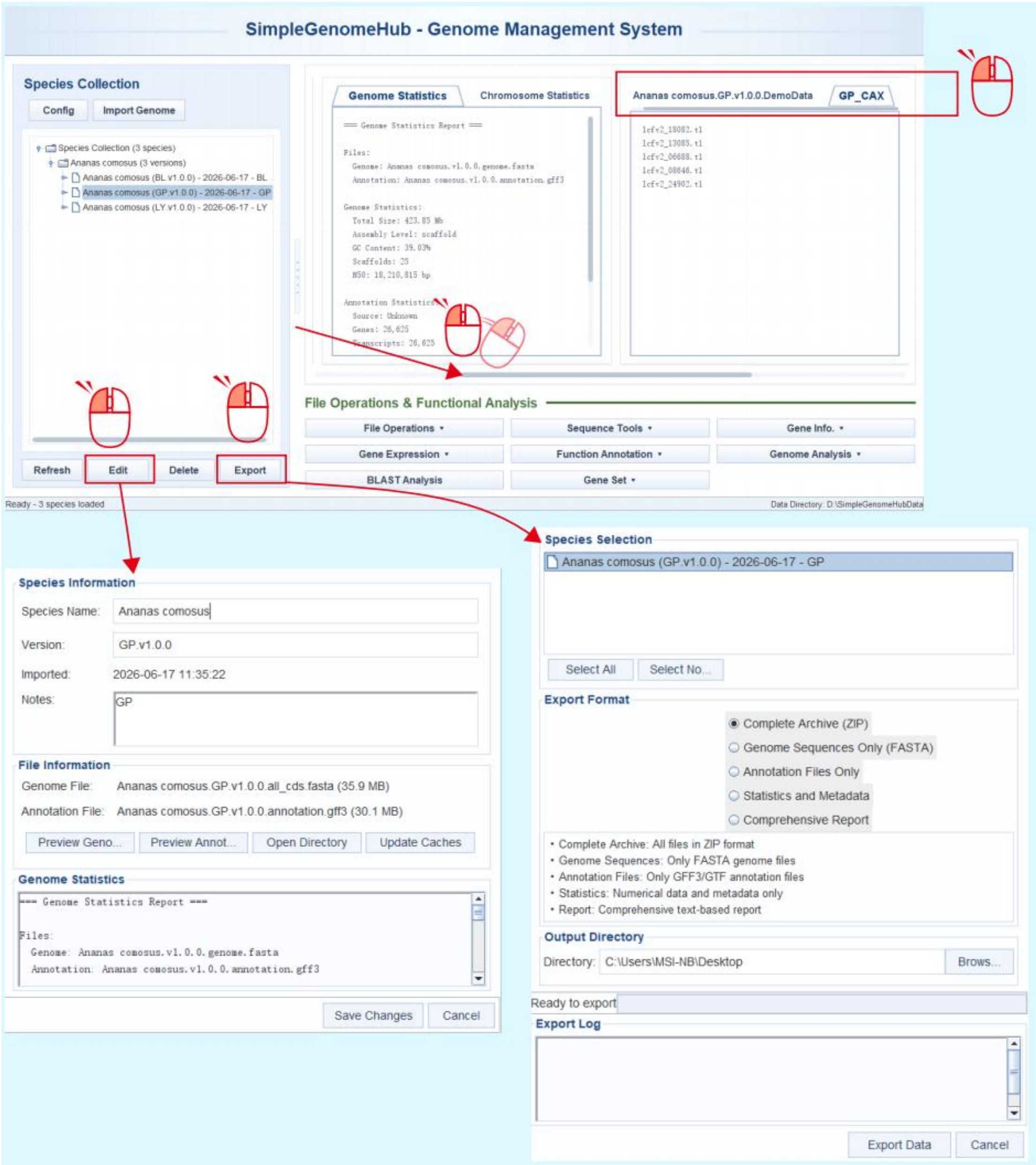


II. Display and Editing of Basic Information for Data Units

- The basic information of the selected data unit is displayed in 5 information sections: **Species Information**, **Genome Statistics**, **Chromosome Statistics, Gene/Region Sets of Interest**, and **Other Data**.
- An **Edit** button is provided for directly editing the genome data unit. Please note that modifying the genome version or name may invalidate related information, such as genome synteny data.
- An **Export** button is provided for one-click packaging and export of the database.






SimpleGenomeHub - Genome Management System
Species Collection
Config
Import Genome
Species Collection (3 species)
Ananas comosus (3 versions)
Ananas comosus (BL v1.0.0) - 2026-06-17 - BL
Ananas comosus (GP v1.0.0) - 2026-06-17 - GP
Ananas comosus (LY v1.0.0) - 2026-06-17 - LY
Genome Statistics
Chromosome Statistics
Ananas comosus.GP.v1.0.0.DemoData
GP_CAX
Refresh
Edit
Delete
Export
File Operations & Functional Analysis
File Operations
Sequence Tools
Gene Info.
Gene Expression
Function Annotation
Genome Analysis
BLAST Analysis
Gene Set
Ready - 3 species loaded
Species Information
Species Name: Ananas comosus
Version: GP.v1.0.0
Imported: 2026-06-17 11:35:22
Notes: GP
File Information
Genome File: Ananas comosus.GP.v1.0.0.all_cds.fasta (35.9 MB)
Annotation File: Ananas comosus.GP.v1.0.0.annotation.gff3 (30.1 MB)
Preview Geno...
Preview Annot...
Open Directory
Update Caches
Genome Statistics
=== Genome Statistics Report ===
Files:
Genome: Ananas comosus.v1.0.0.genome.fasta
Annotation: Ananas comosus.v1.0.0.annotation.gff3
Save Changes
Cancel
Species Selection
Ananas comosus (GP.v1.0.0) - 2026-06-17 - GP
Select All
Select No...
Export Format
Complete Archive (ZIP)
Genome Sequences Only (FASTA)
Annotation Files Only
Statistics and Metadata
Comprehensive Report
• Complete Archive: All files in ZIP format
• Genome Sequences: Only FASTA genome files
• Annotation Files: Only GFF3/GTF annotation files
• Statistics: Numerical data and metadata only
• Report: Comprehensive text-based report
Output Directory
Directory: C:\Users\MSI-NB\Desktop
Brows...
Ready to export
Export Log
Export Data
Cancel

I. Flexible Use of Gene Sets

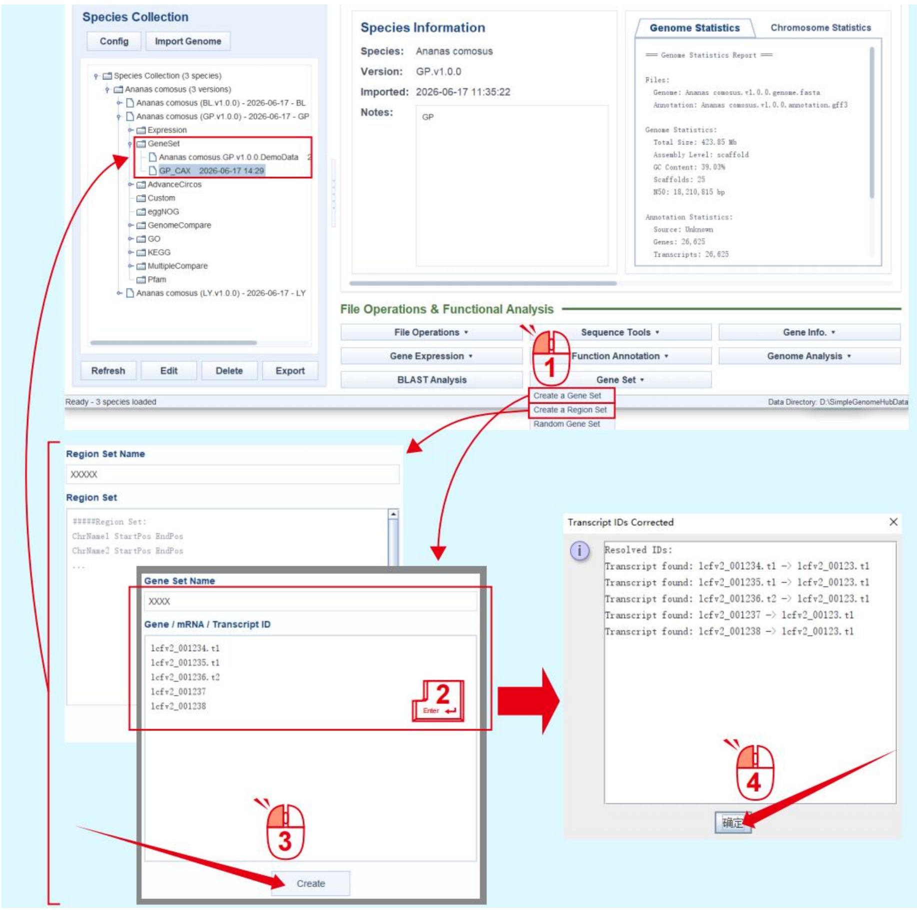
In studies of biological traits, research objectives and applications are almost always ultimately linked to genes. However, during the research cycle from data mining to publication, it is important to properly preserve the genes and genomic regions of interest identified through analysis. For this purpose, Simple Genome Hub provides a Gene Set function, supporting two types of saved collections: gene sets and region sets.

1. After selecting a data unit, click **Create a Gene Set** / **Create a Region Set** in the **Gene Set** section.

2. Edit the name of the gene/region set and enter the corresponding genomic positions or gene IDs. Gene sets are saved in the form of transcript IDs. During input, the software automatically checks whether the entered IDs match the selected genome. If only a gene ID is entered, all transcripts under that gene will be automatically added.

3. Click the **Create** button. The software first performs a matching check and then displays the matching result in a pop-up window.

4. Click the **confirmation** button to complete creation. Newly added gene sets and region sets are automatically deployed under the GeneSet directory of the corresponding genome data unit.



Gene sets are accessible through interfaces in almost all analytical functions and can be used directly as analysis targets or as intermediate inputs in functional workflows. To facilitate functional exploration, after a new genome data unit is imported, Simple Genome Hub automatically generates a random gene set containing 100 random transcript IDs. This gene set is named using the genome name followed by DemoData. Users can directly select this gene set as input when testing new functions.

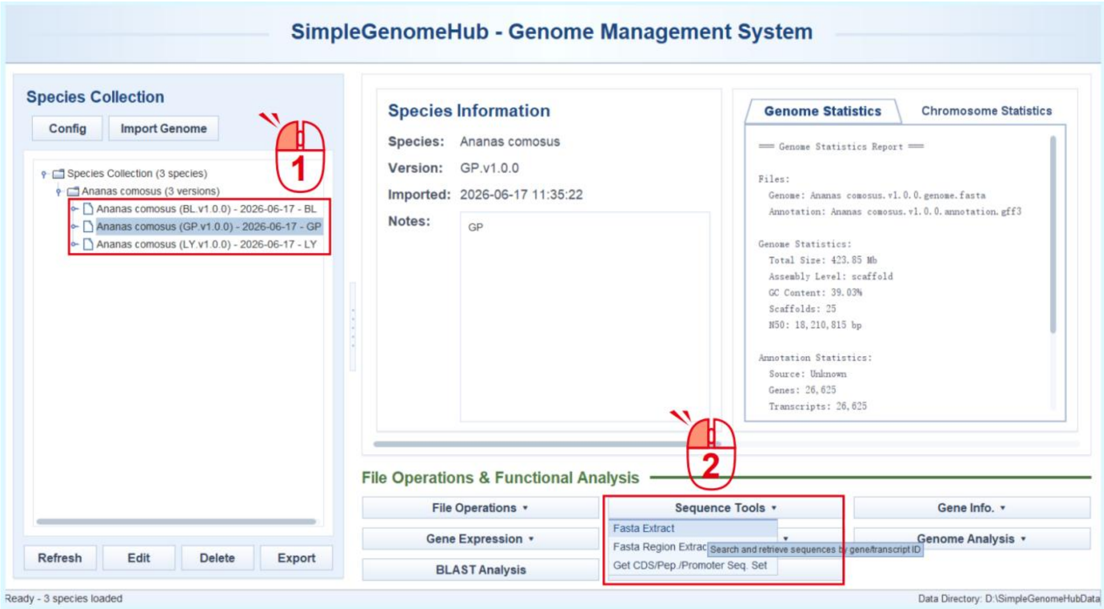
The Gene Set section also provides a Random Gene Set function, which can be used to regenerate or update the random gene set for the current genome data unit.

Simple Genome Hub | 4. Sequence Operations

I. Extract FASTA Sequences by Gene ID

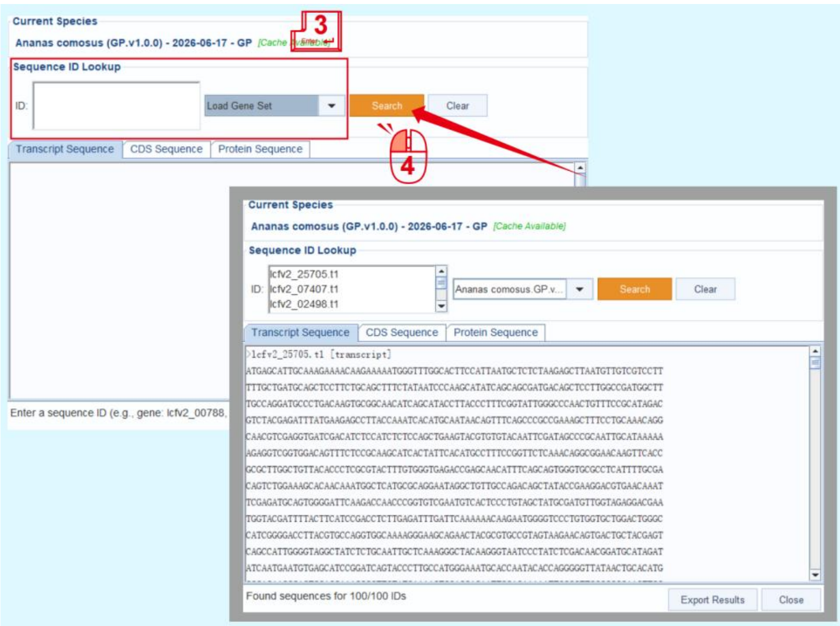
1. Select the data unit from which sequences will be extracted.

2. Click **Fasta Extract** under **Sequence Tools**.



3. Enter the gene/transcript IDs to be extracted in the text box, or select an existing gene set from the drop-down menu under **Load Gene Set** on the right.

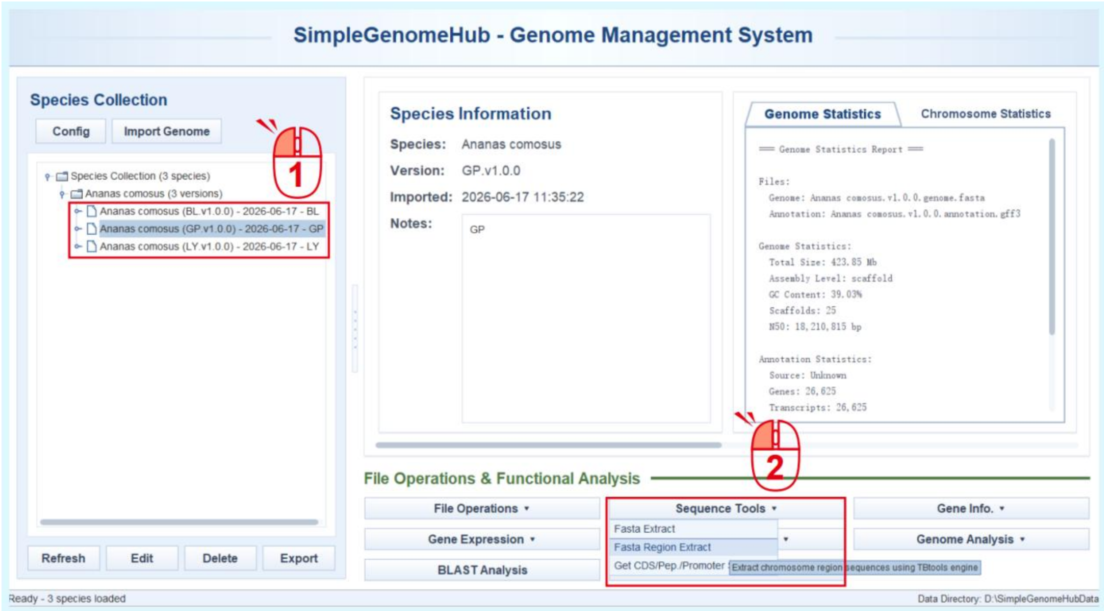
4. Click **Search** button to complete the extraction. The extracted sequences will be displayed directly in the text box below. You can switch among **Transcript Sequence**, **CDS Sequence**, and **Protein Sequence** to display different sequence types.



II. Extract FASTA Sequences from Target Regions

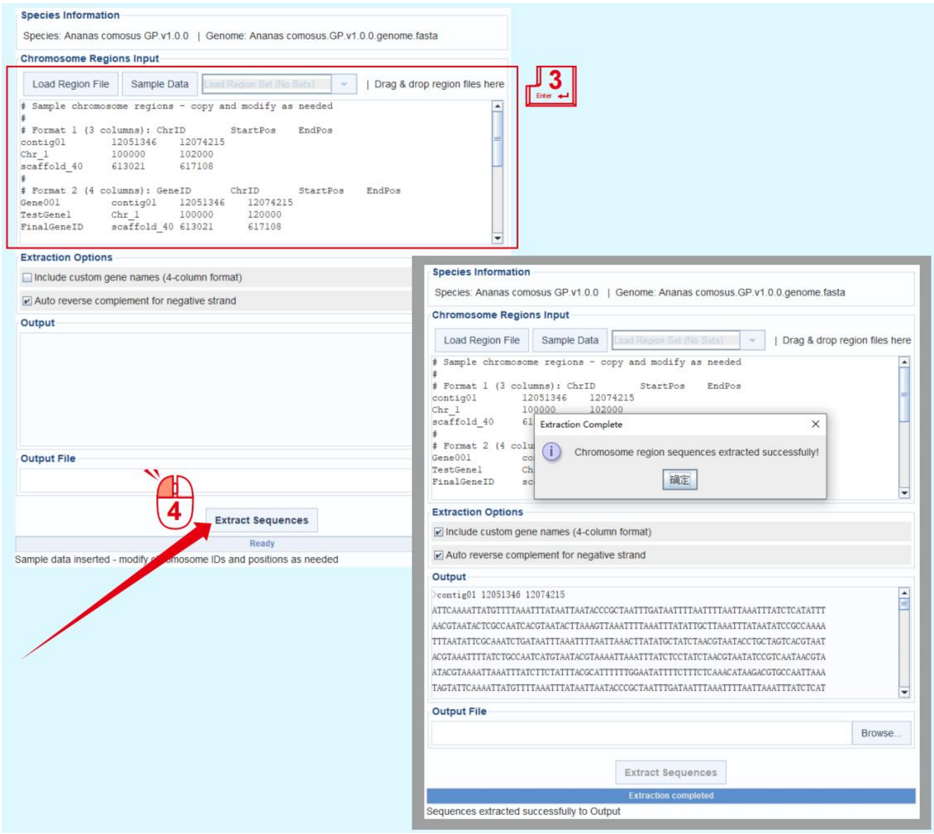
1. Select the data unit from which sequences will be extracted.

2. Click **Fasta Region Extract** under **Sequence Tools**.



3. Enter the target region in the text box according to the instructions, or select an existing region set from the drop-down menu under **Load Region Set** at the top.

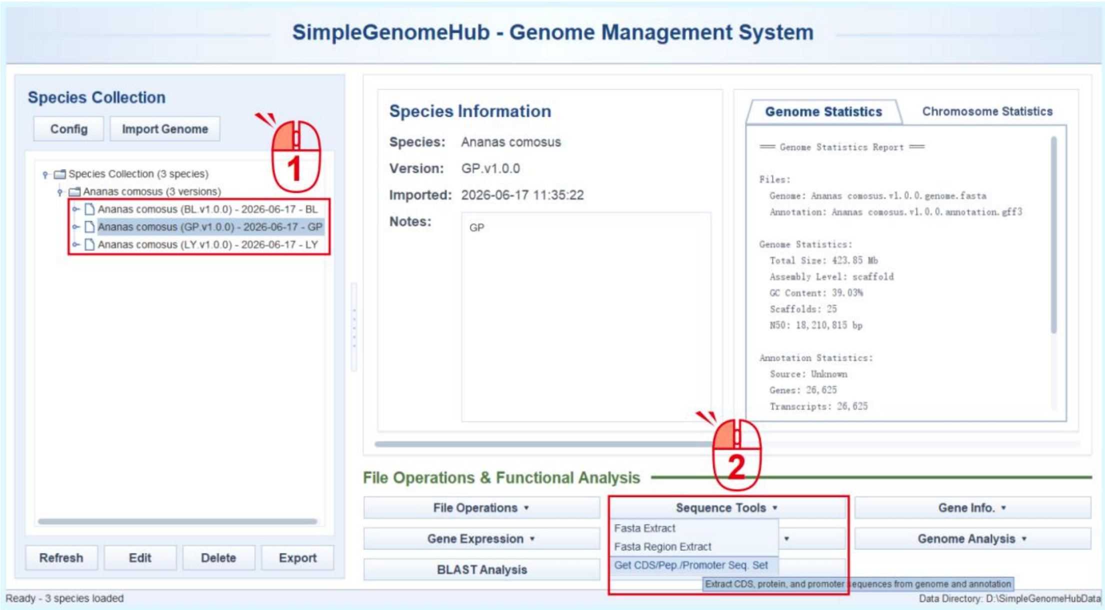
4. Click the **Extract Sequences** button to complete the extraction.



II. Extract Upstream Gene Sequences

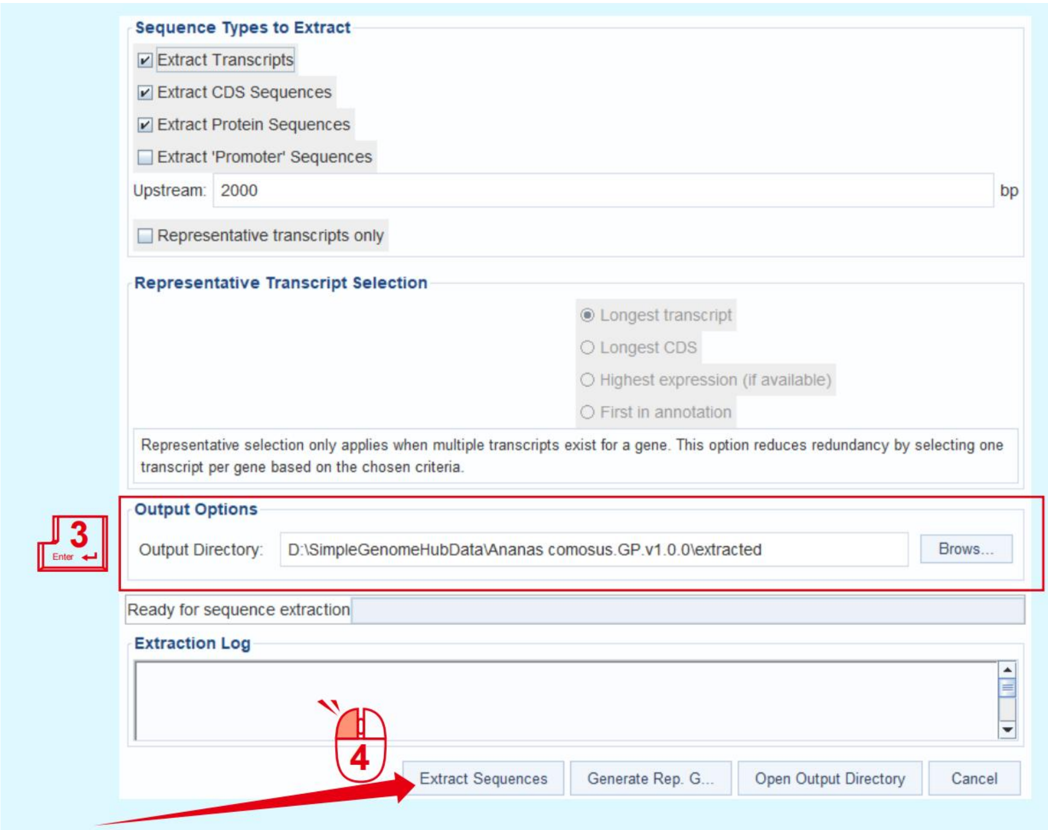
1. Select the data unit from which sequences will be extracted.

2. Click **Get CDS/Pep/Promoter Seq. Set** under **Sequence Tools**.



3. Edit or select the export location under **Output Options**.

4. Click the **Extract Sequences** button to complete the extraction of upstream / promoter sequences based on the annotation information of the selected data unit.

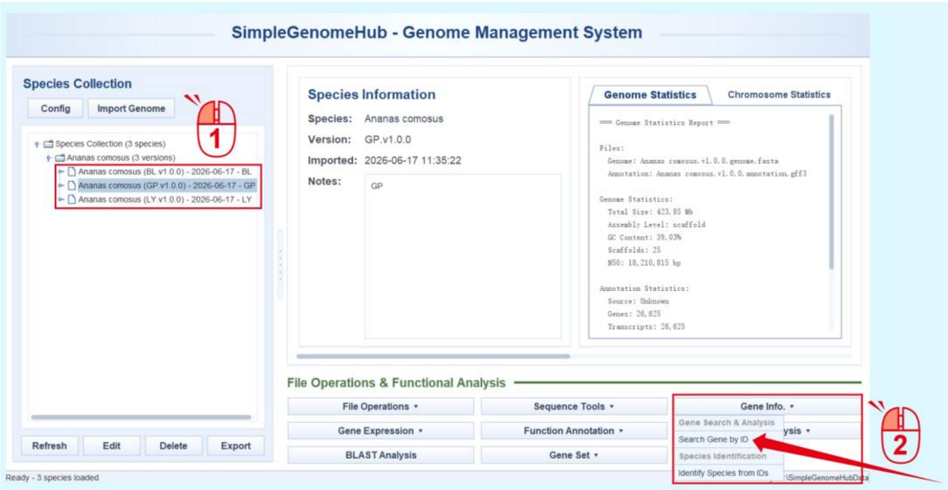


Simple Genome Hub | 5. Gene ID Search and Reverse Tracing

I. Query All Related Information by Gene ID

1. A genome data unit must be selected first (Since different genome versions may use similar ID naming systems, a global search function is not provided), and the query is then performed within the selected genome.

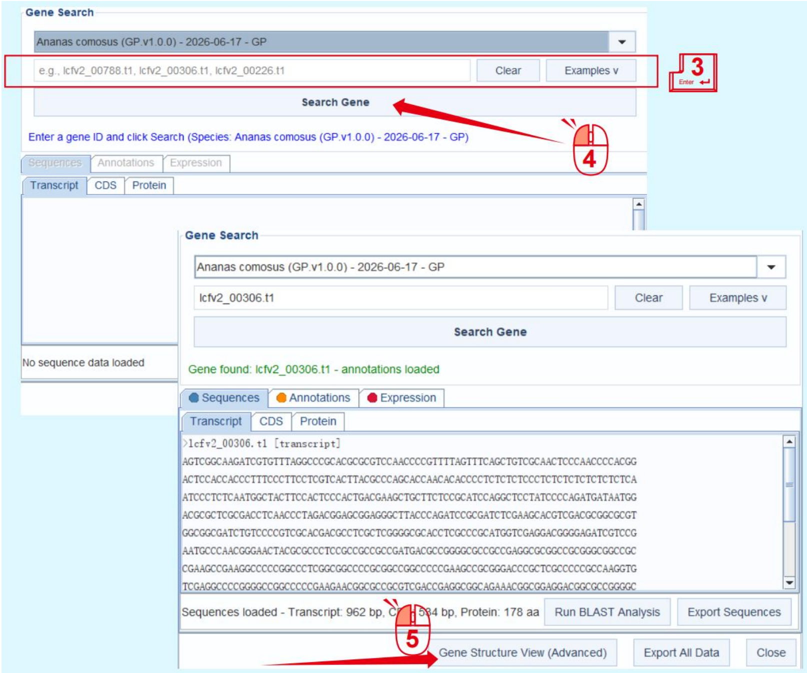
2. Click **Search Gene by ID** under **Gene Info**.

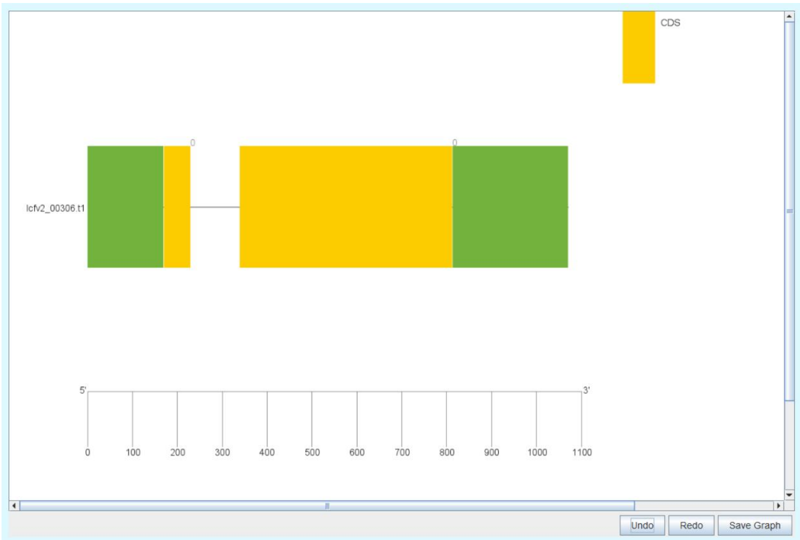


3. Enter the gene / transcript ID to be queried in the text box. If you do not remember the naming format of the annotated genes, click **Examples** to automatically fill in a random example ID.

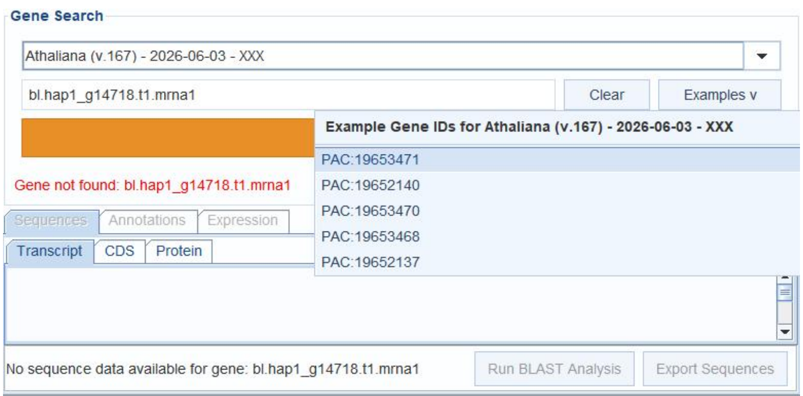
4. Click the **Search Gene** button to perform the query. The query results include three ID-related display tabs: **Sequences**, **Annotations**, and **Expression**. The latter two are available only when the corresponding data have been imported. You can switch among these tabs by clicking them. If gene functional annotation information has already been imported, loading GO annotations may take some time because GO files are relatively slow to load.

5. Click the **Gene Structure View (Advanced)** button to visualize the selected gene using the Gene Structure View (Advanced) function in TBtools-II.





- If you are unsure about the transcript ID format used in the current genome annotation, click **Examples** to view example IDs from the current genome. A warning will be displayed if an ID in an incorrect format is entered:

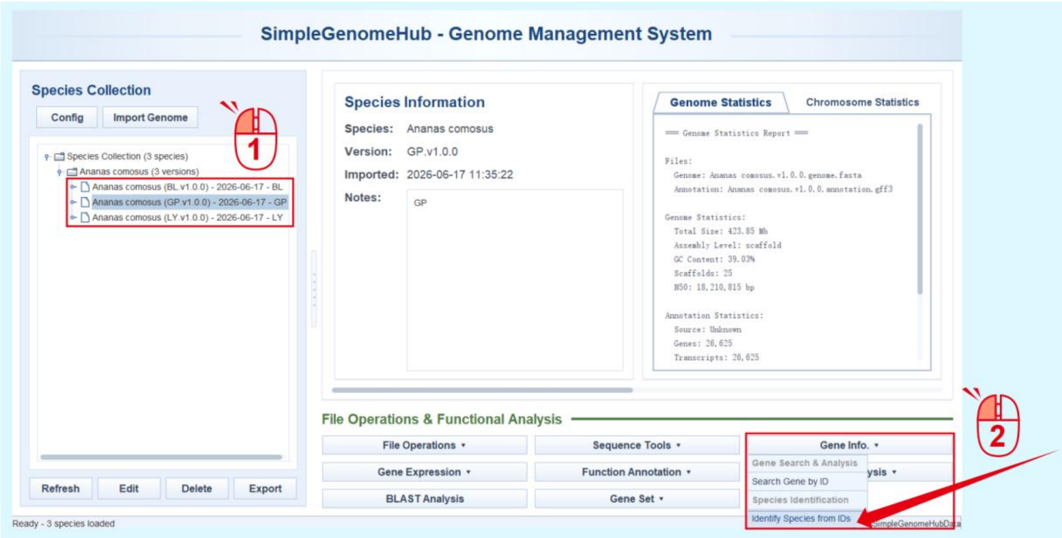


II. Identify the Species and Genome Based on Known IDs

During analysis, researchers may forget the source of a gene of interest due to a long analysis cycle or inconsistent data management. This function can be used to accurately trace the origin of a known ID.

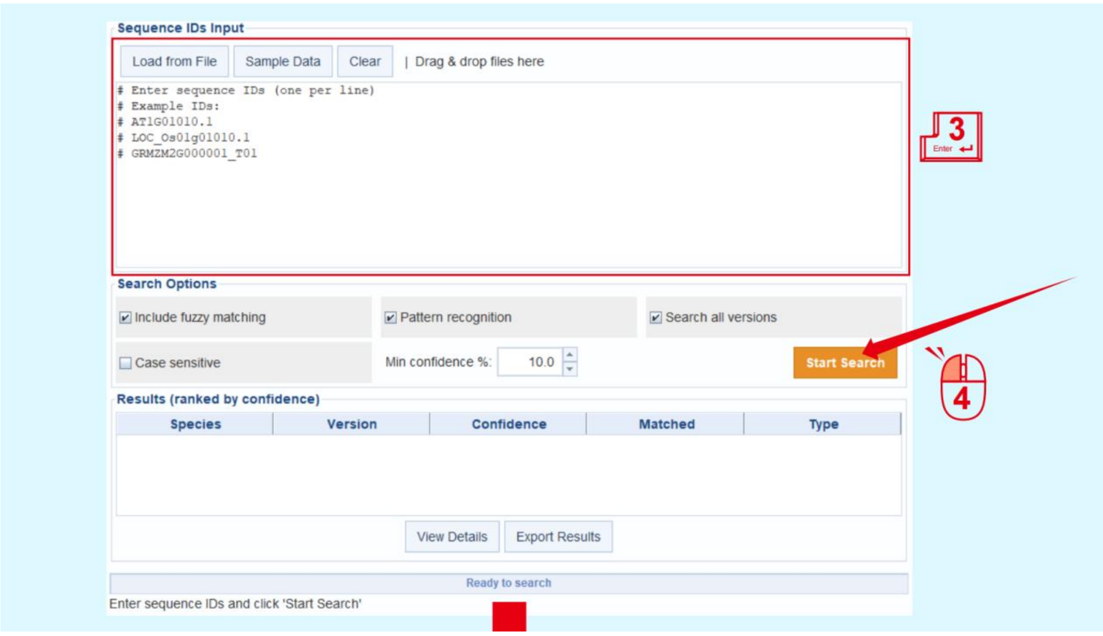
1. Click any data unit. The selected data unit does not affect the search.

2. Click **Identify Species from IDs** under **Gene Info**.

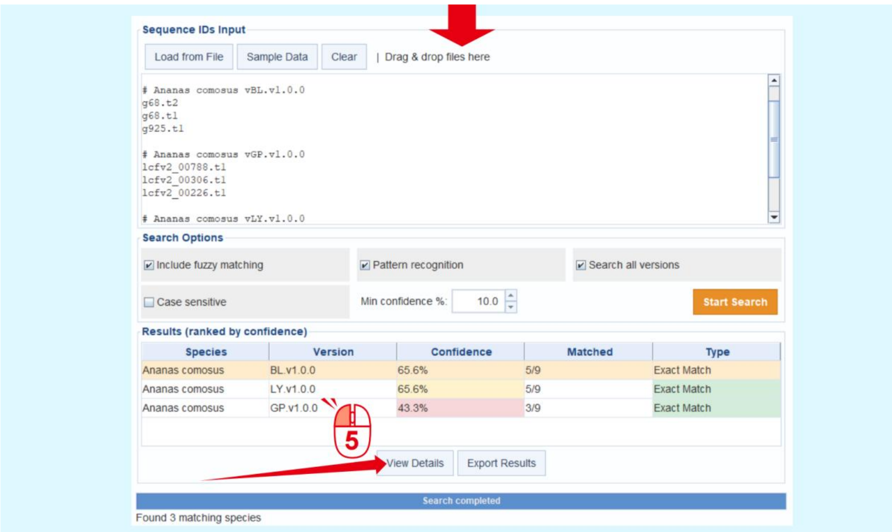


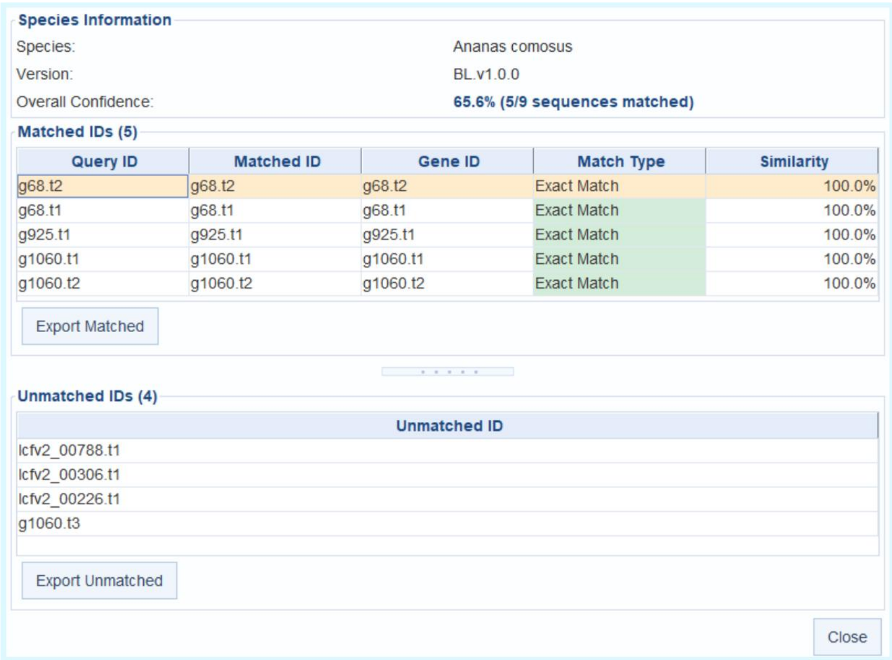
3. Enter the ID to be queried in the text box.

4. Click **Start Search** to perform the query. The results will be listed directly under **Results**, together with information such as matching scores.

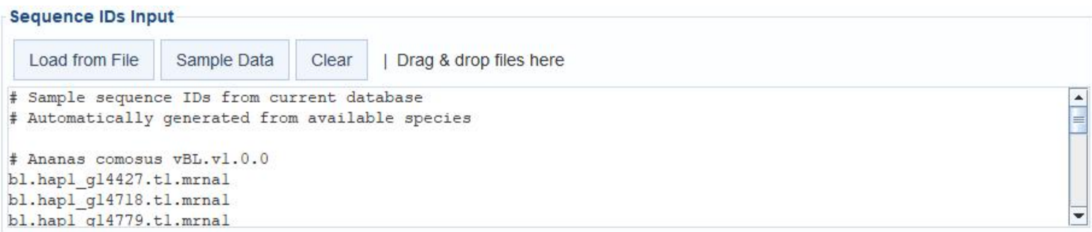


5. Click **View Details** to inspect the detailed search results.





- During long-term data management, users may forget the annotation ID format used for each genome. In this case, click **Sample Data** to automatically generate example IDs for all imported genomes as a reference:



Simple Genome Hub | 6. BLAST Analysis

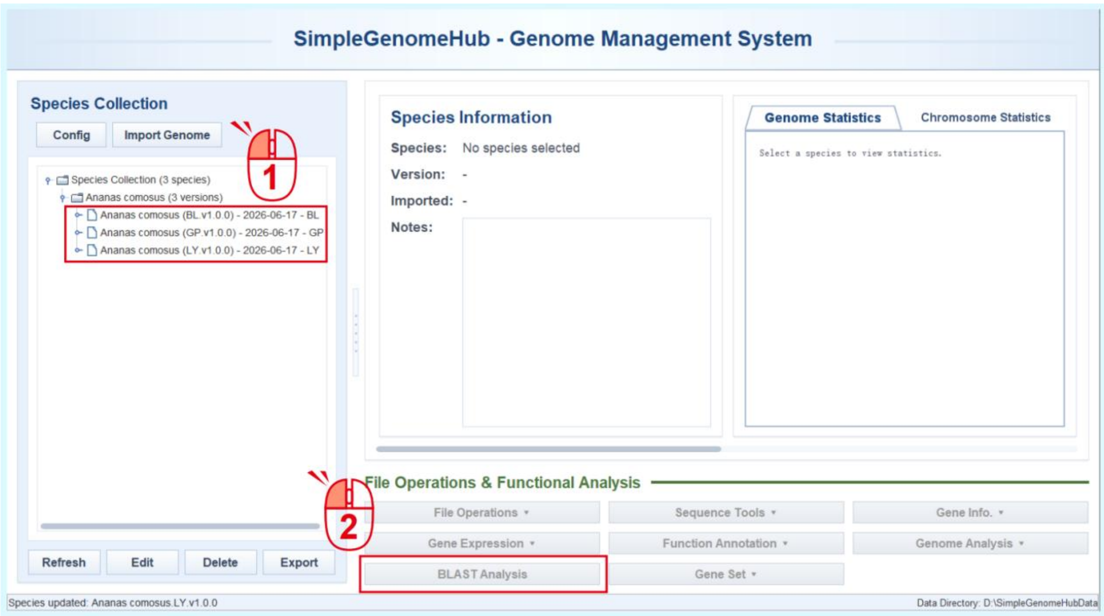
Sequence similarity search for homologous genes is an essential workflow in genome data analysis and mining. BLAST-based searches may involve within-genome comparisons, external sequence comparisons, or batch comparisons. The target databases may include genomic sequences, gene sequences, transcript sequences, protein sequences, and other sequence types.

In Simple Genome Hub, the BLAST Analysis function integrates all of these search requirements into a single interface. The results can be viewed and saved directly within the same interface. In addition, BLAST entry points are provided in functions such as Gene Set and Gene ID Search, allowing users to directly jump to BLAST analysis from related workflows.

I. Standard Workflow

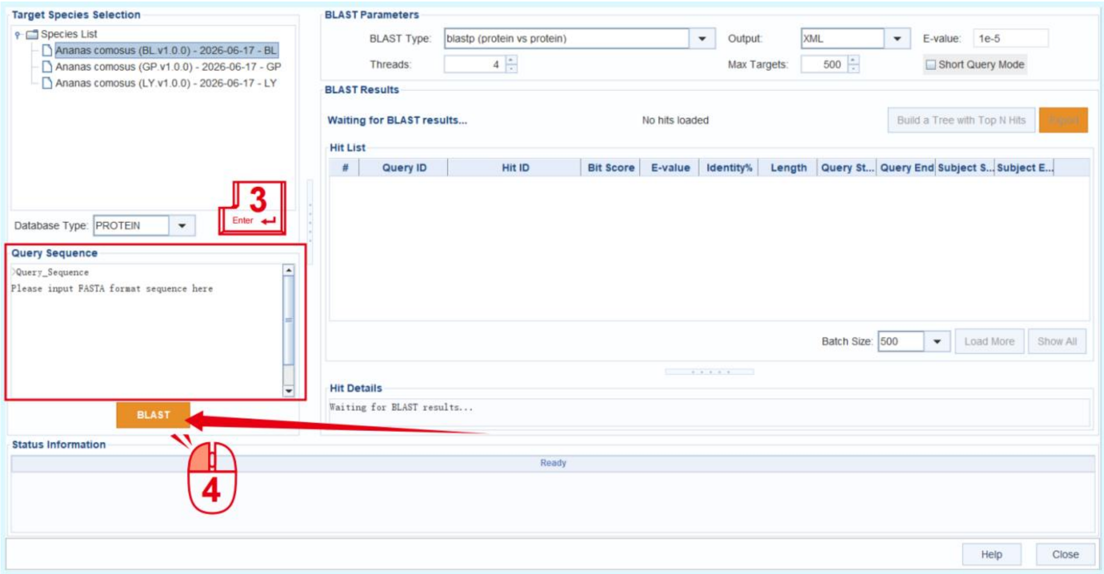
1. Select the genome data unit for BLAST analysis.

2. Click the **BLAST Analysis** button.

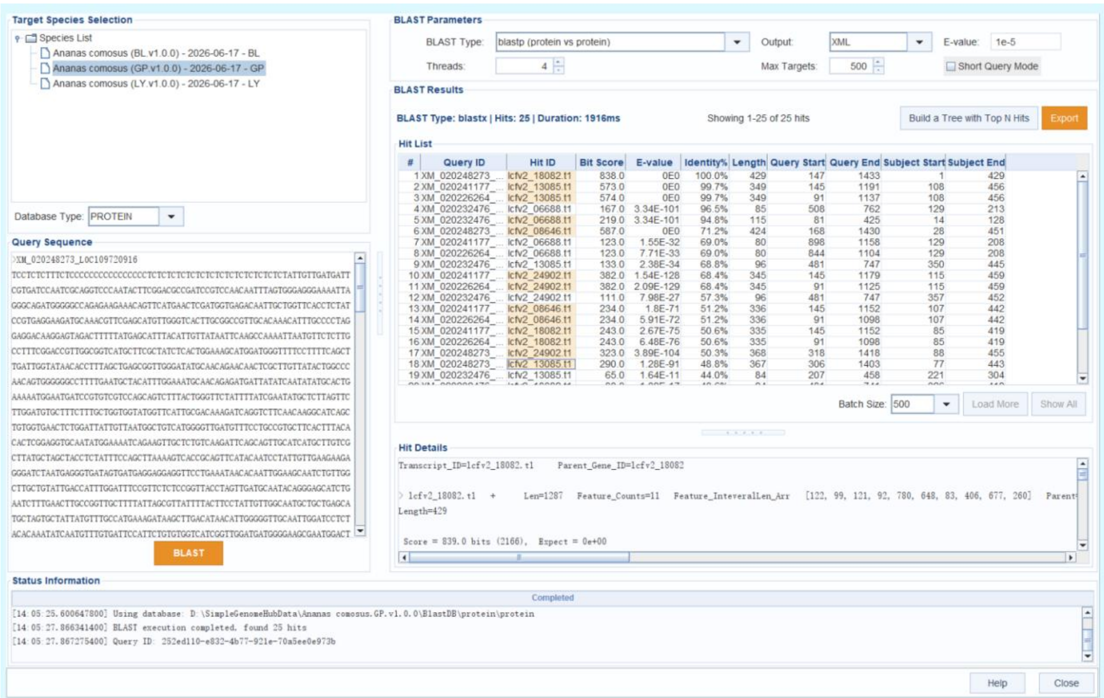


3. Paste the query sequence into the input box. The interface automatically determines the appropriate BLAST Type based on the input sequence and the selected target database.

4. Click the **BLAST** button to run the search.



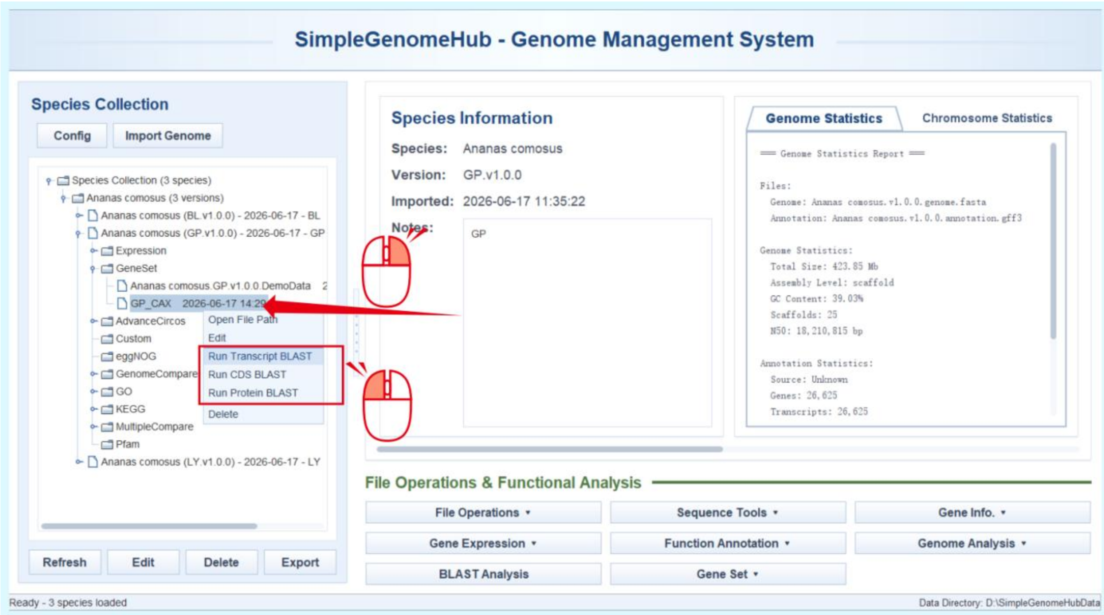
The BLAST results are displayed directly in the BLAST Results panel on the right in table format. The table supports sorting, duplicate removal, free selection, copying, and other operations. In the result table, users can right-click selected genes and use the Gene Set function to create a gene set from the BLAST results.



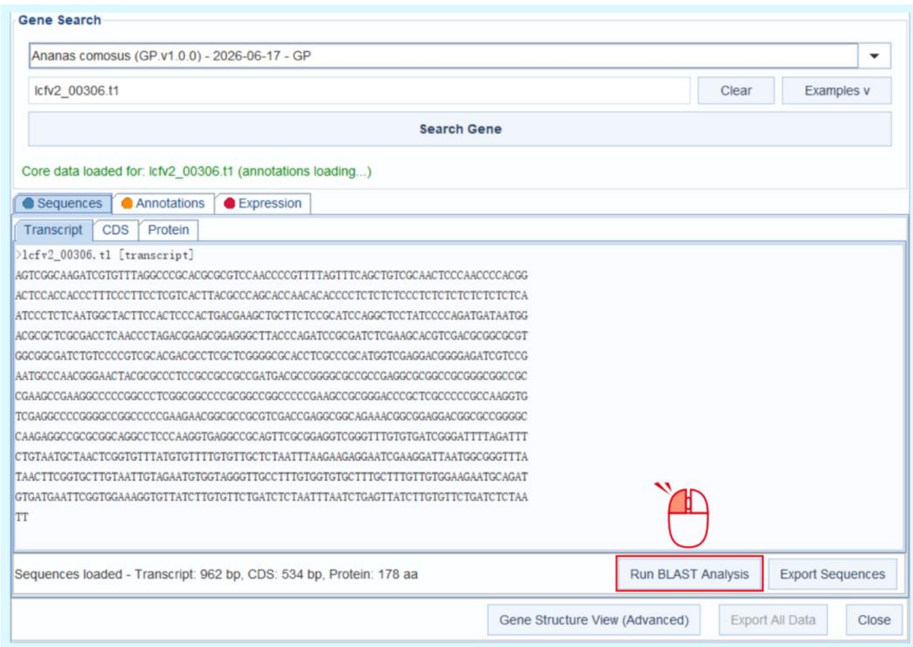
II. Alternative Entry Points

To facilitate rapid sequence similarity searches for genes of interest, shortcut options are also provided in the right-click menu of GeneSet files on the main interface and in the Search Gene by ID interface. After jumping to the BLAST interface, the corresponding sequence is automatically filled into the input box, and users can directly click the BLAST button to perform the search.

- Right-click a GeneSet file and select **Run BLAST** from the pop-up menu.



- After searching by gene ID, users can also select different sequence types, such as CDS or Protein, and directly jump to BLAST analysis.

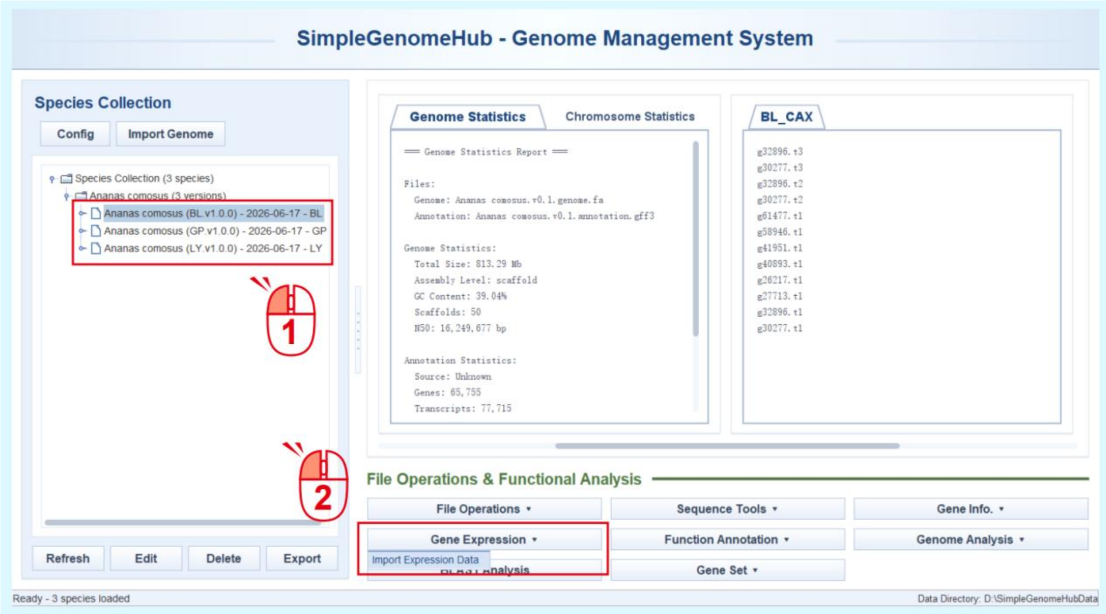


Simple Genome Hub | 7. Expression Data Import and Exploration

I. Import Expression Data

1. Select the genome data unit into which expression data will be imported.

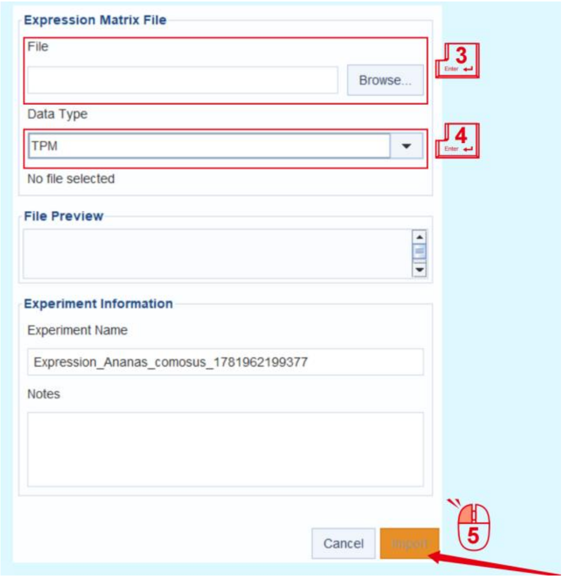
2. Click **Import Expression Data** under **Gene Expression**.



3. Select the expression data file to be imported.

4. Check the file type carefully.

5. Click **Import** button. The software will first perform matching and error checking.

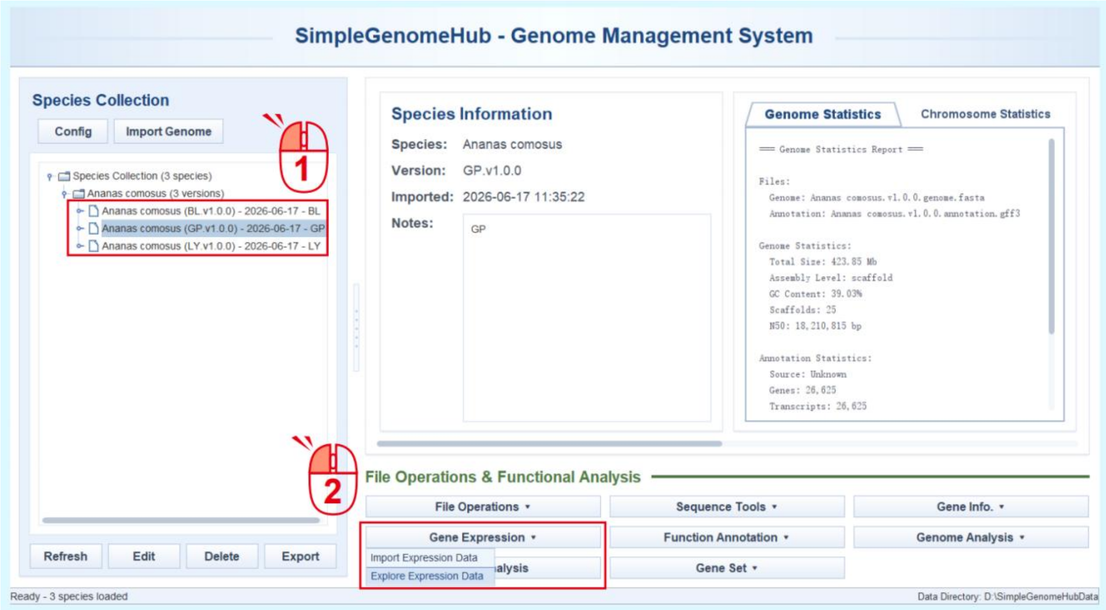


Check the file type car

II. Explore Expression Data

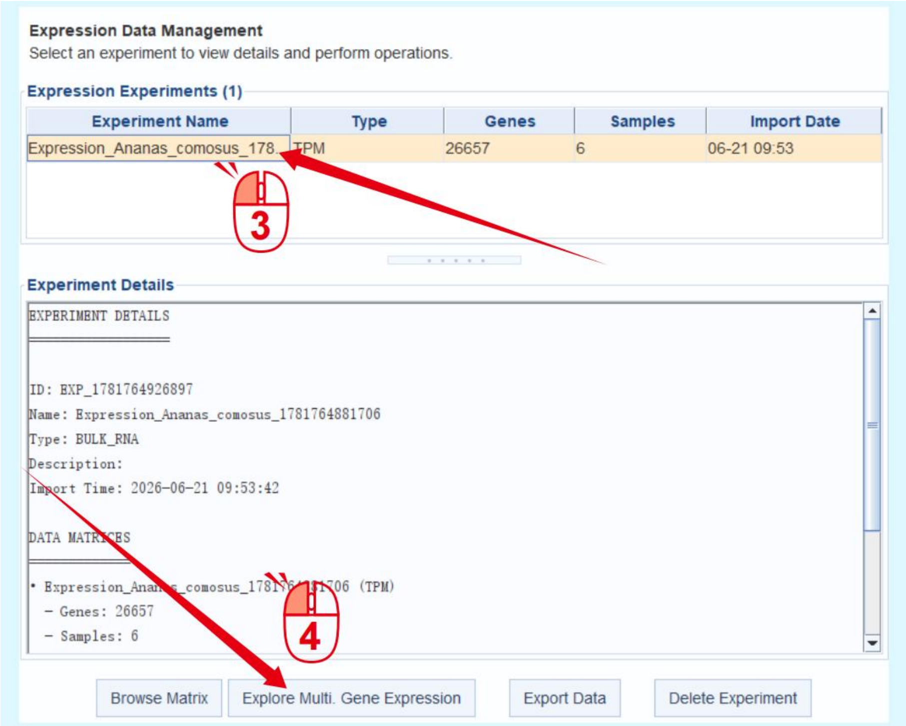
1. Select the genome data unit for which expression data will be explored.

2. Once expression data have been imported for the selected genome data unit, an **Explore Expression Data** option will appear under **Gene Expression**.



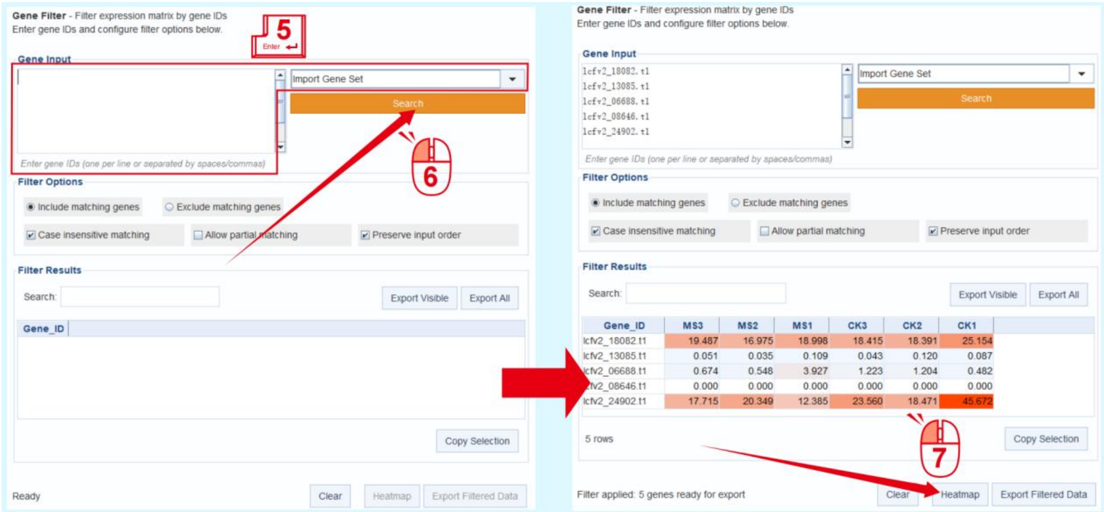
3. In the pop-up interface, select the imported expression dataset.

4. Click **Explore Multi. Gene Expression** to perform joint expression analysis of multiple genes.

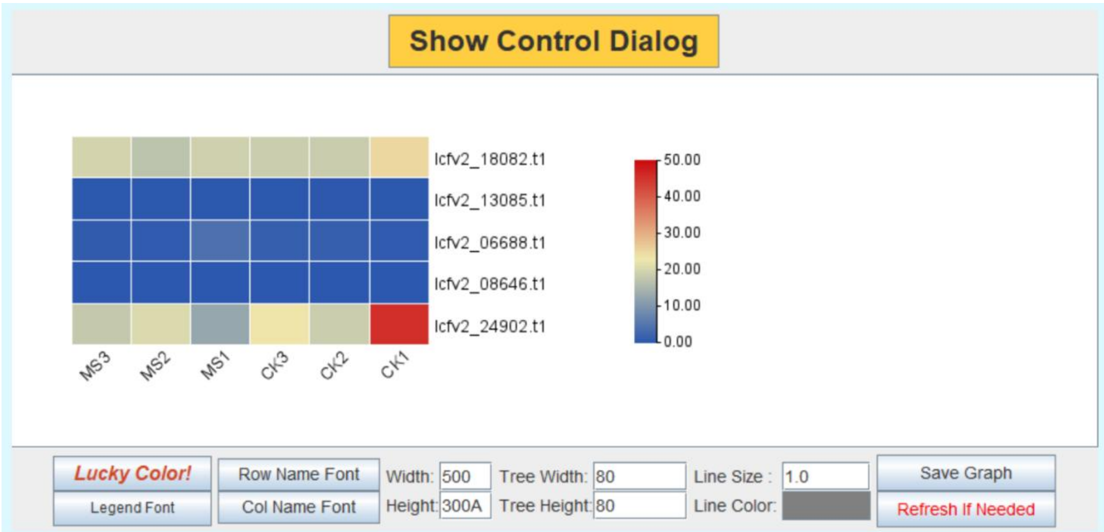


5. Enter gene IDs manually or select a gene set to fill them in directly.

6. Click **Search** to start extraction. The extracted results will be displayed directly under **Filter Results** as a heatmap-style table, and the data can be freely rearranged.



7. Click the **Heatmap** button to directly invoke the Heatmap function of TBtools-II for visualization.



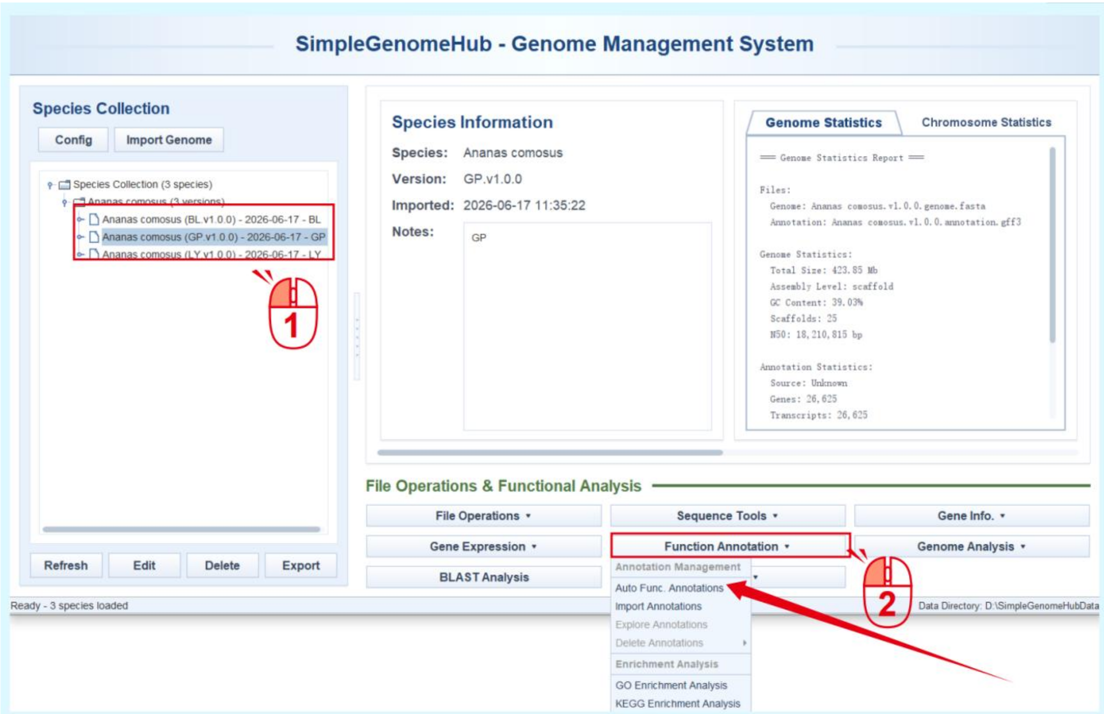
Simple Genome Hub | 8. Gene Functional Annotation and Enrichment Analysis

Gene functional annotation files can be obtained in two ways: automatically generated by Simple Genome Hub or manually imported by the user. Manual import follows a workflow similar to expression data import and can be accessed through Import Annotations under Function Annotation. This section only introduces the automatic annotation workflow.

I. Automated Gene Functional Annotation

1. Select the genome data unit for functional annotation.

2. Click **Import Annotations** under **Function Annotation** to open the annotation interface.



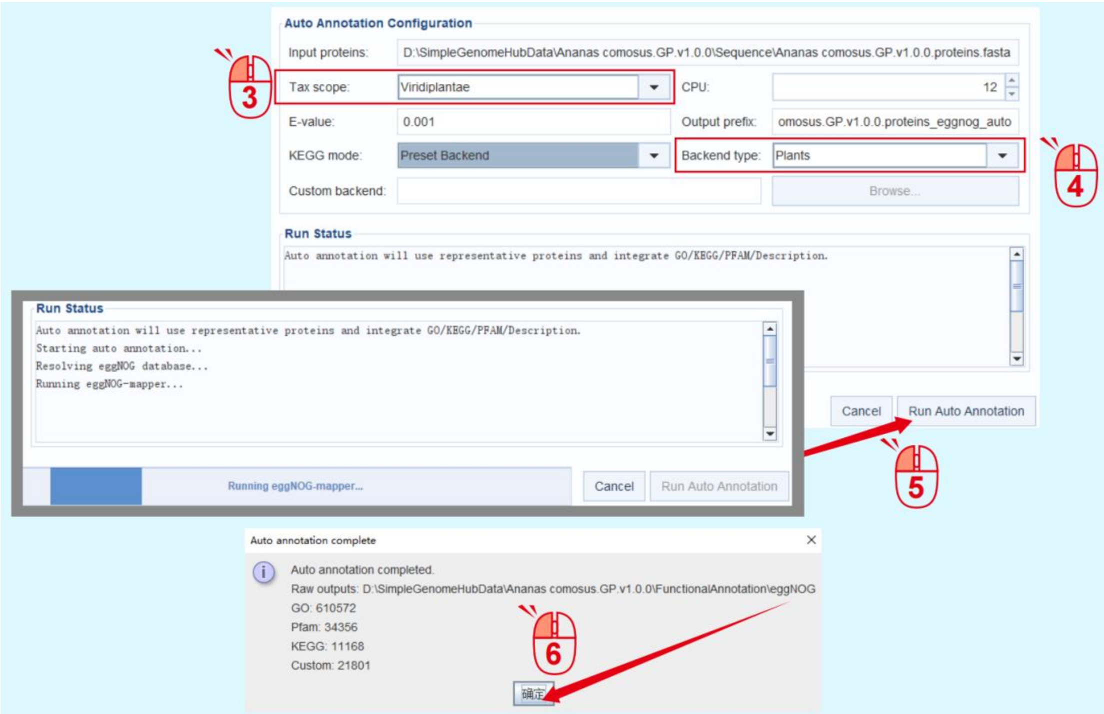
Functional annotation is performed using a Java-rewritten version of eggNOG-mapper. The workflow follows the original eggNOG-mapper procedure; therefore, the following parameters need to be properly selected:

3. Tax scope；

4. Backend type；

5. After confirming that the parameters are correct, click **Run Auto Annotation** to start automated annotation. Simple Genome Hub will automatically download the selected database and perform functional annotation.

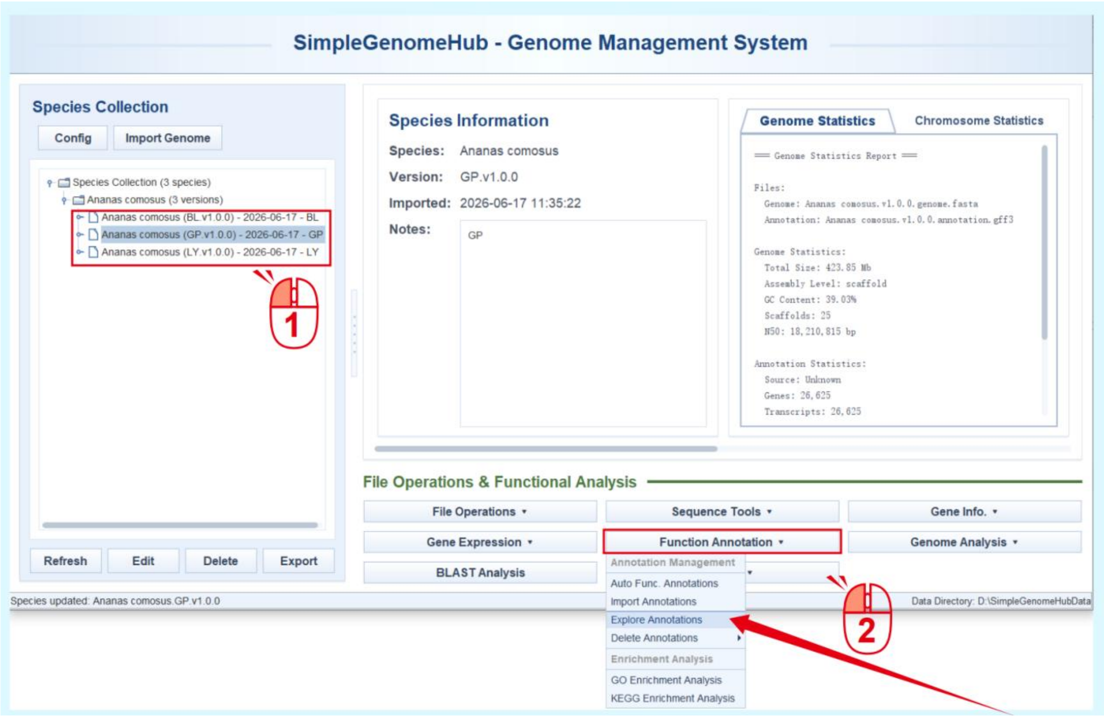
6. After annotation is completed, a pop-up message will be displayed. Click the **confirmation** button to complete the process.



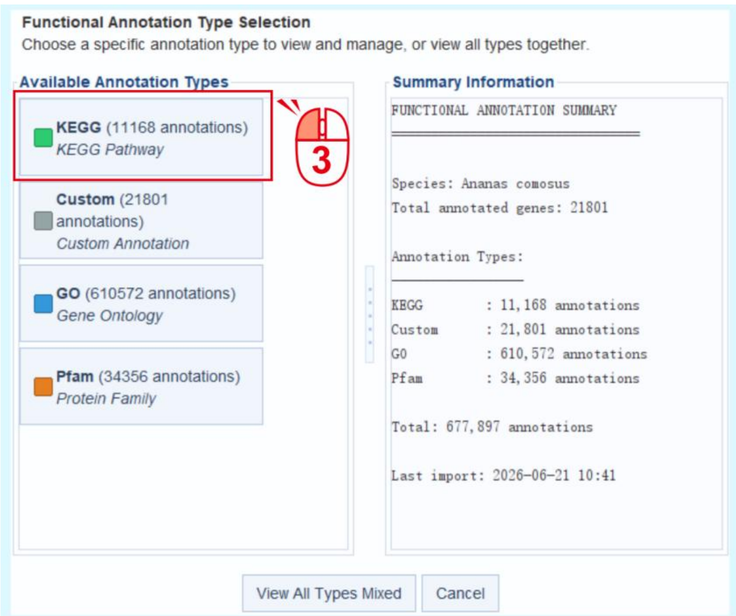
II. Query Gene Functional Annotations

1. Select a data unit.

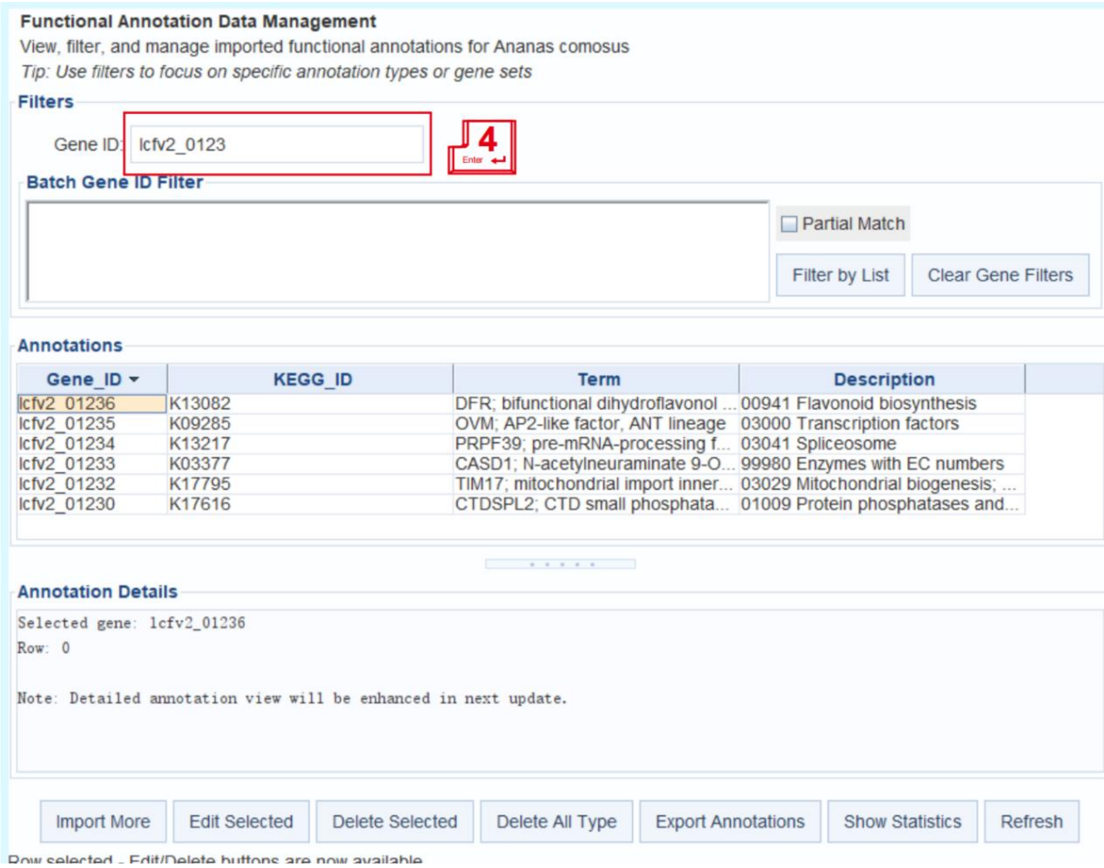
2. After gene functional annotation files are available, the related functions under **Function Annotation** become selectable. Here, click **Explore Annotations**.



3. In the pop-up interface, the available functional annotation types are listed as cards on the left. Here, **KEGG** is used as an example for annotation query and exploration.



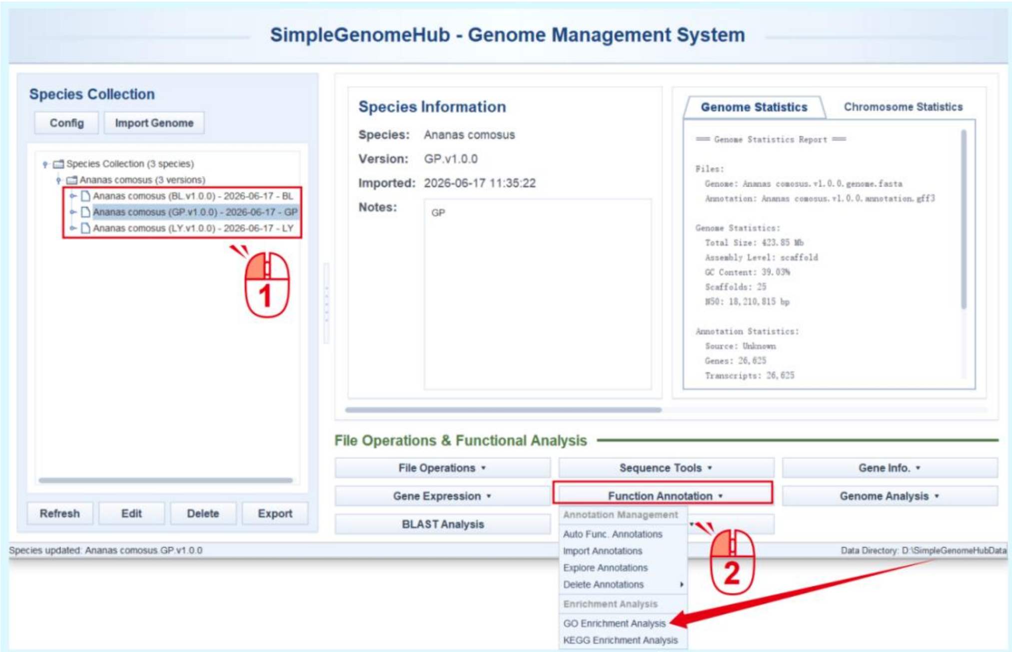
4. In the newly opened interface, enter the ID to be queried in the input box at the top. The Annotations list below will update the search results in real time according to the input.



II. GO / KEGG Enrichment Analysis

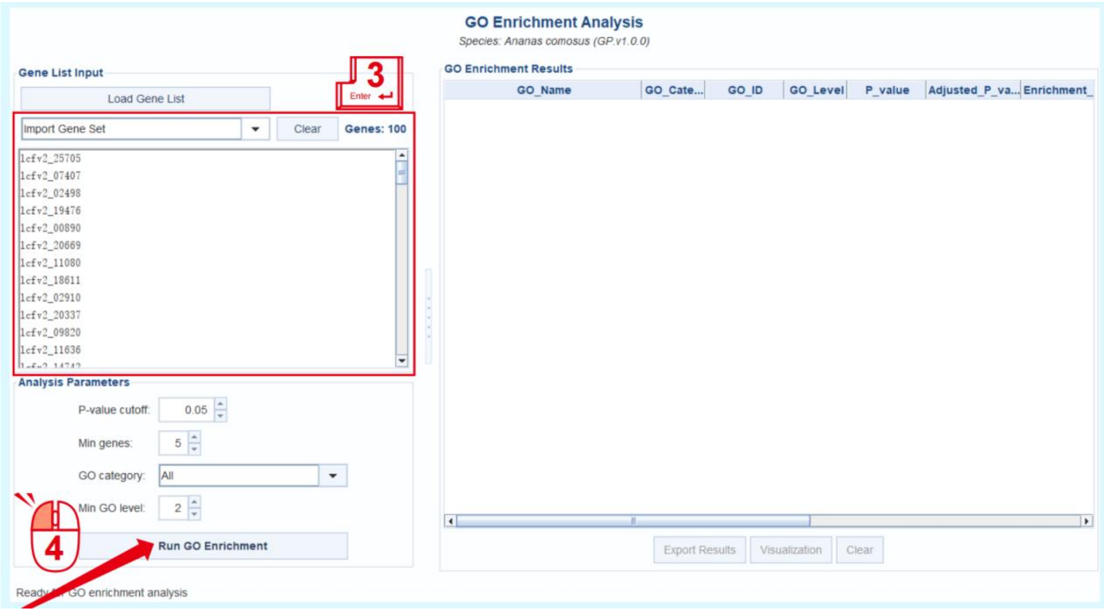
1. Select a data unit.

2. Select **GO Enrichment Analysis** under **Function Annotation**. The workflow for KEGG enrichment analysis is the same as that for GO enrichment analysis.

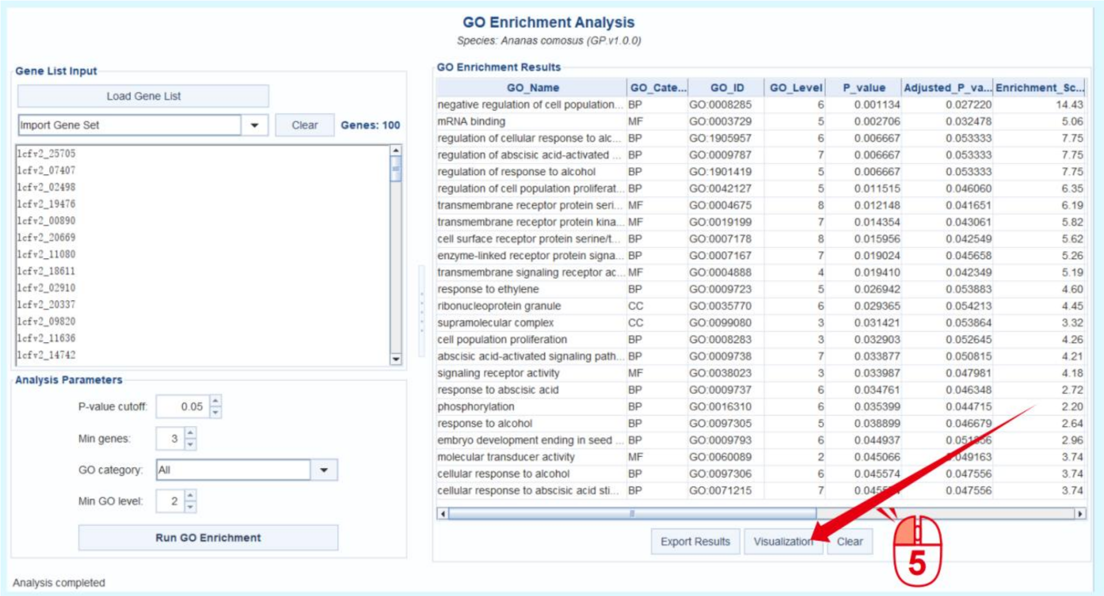


3. Enter the genes for enrichment analysis on the left, or select an existing gene set from the drop-down menu to fill in the input directly.

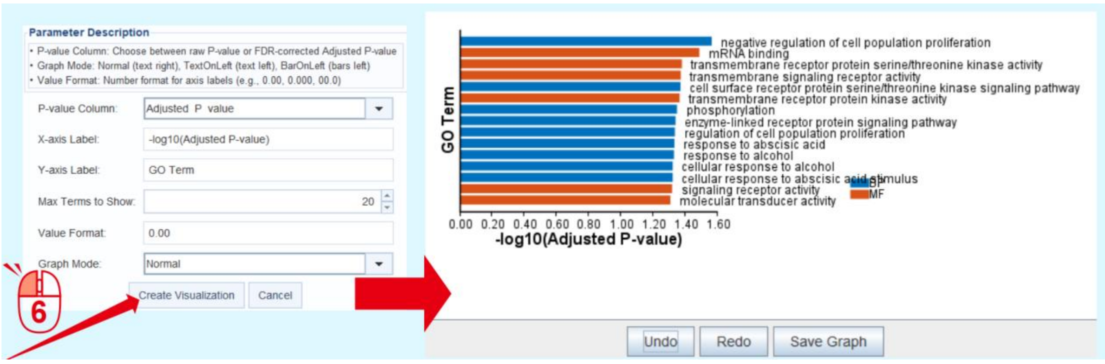
4. Click **Run GO Enrichment** to perform enrichment analysis. The results will be listed directly on the right.



5. Click **Visualization** below the results to visualize the enrichment analysis output.



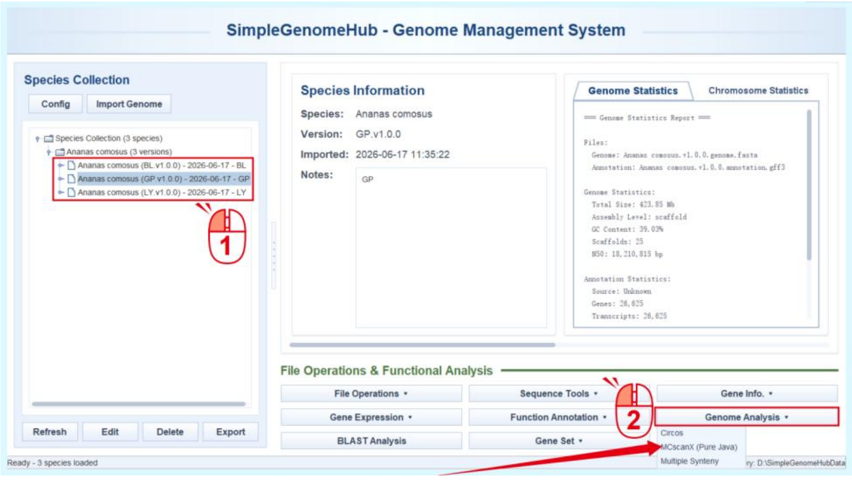
6. Before visualization, a pop-up window will display the enrichment-related parameters. In most cases, the default settings can be used. Click **Create Visualization** to generate the visualization results. The visualization is rendered by invoking the GO Enrichment function of TBtools-II.



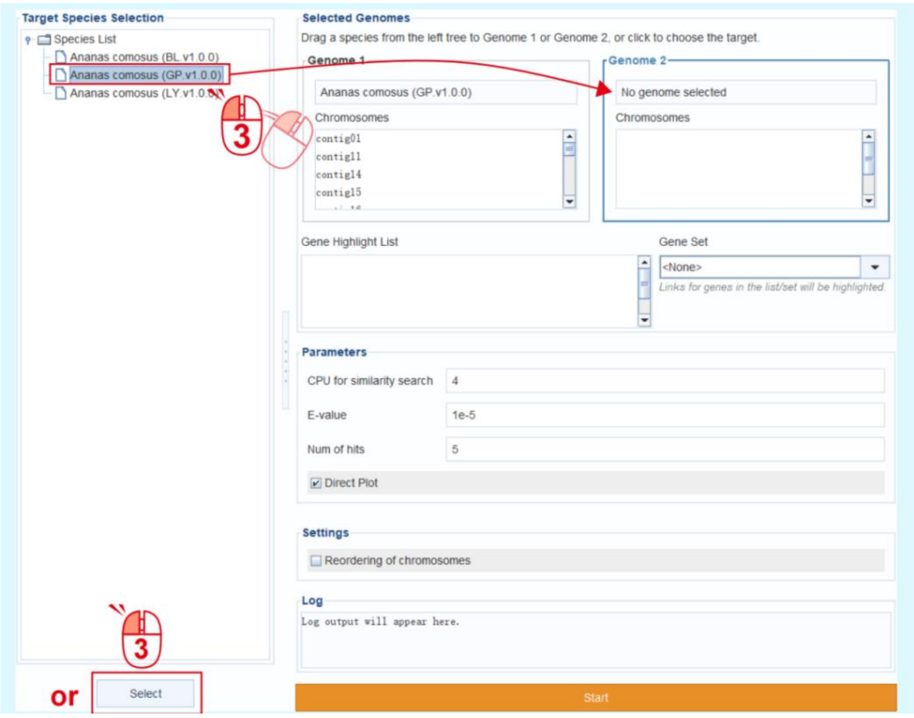
Simple Genome Hub | 9. Pairwise Genome Synteny Analysis

1. Select one of the genome data units to be used for synteny analysis. The two genomes to be compared can be changed later, but the related results will be saved under the genome data unit selected at this step.

2. Click **MCscanX (Pure Java)** under **Genome Analysis**.

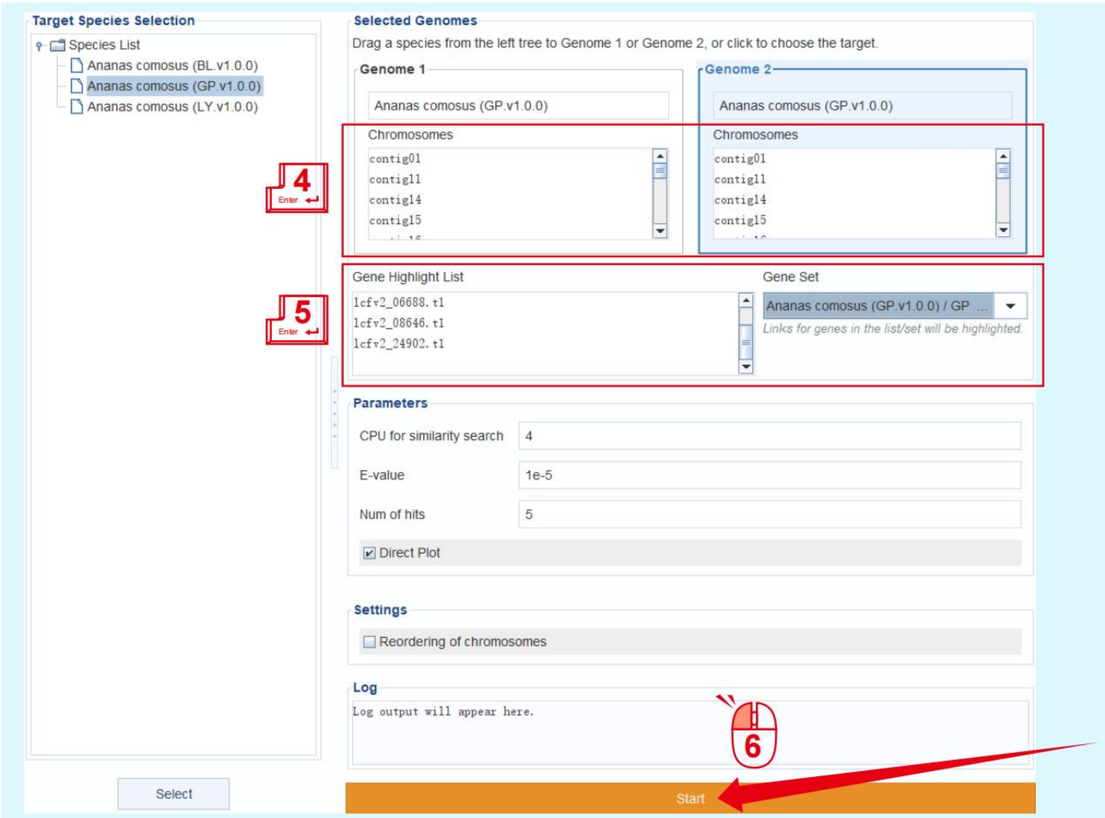


3. In the pop-up interface, the second genome for synteny analysis can be directly dragged into **Genome 2**, or selected by clicking **Select** below. In this example, GP is compared with itself.

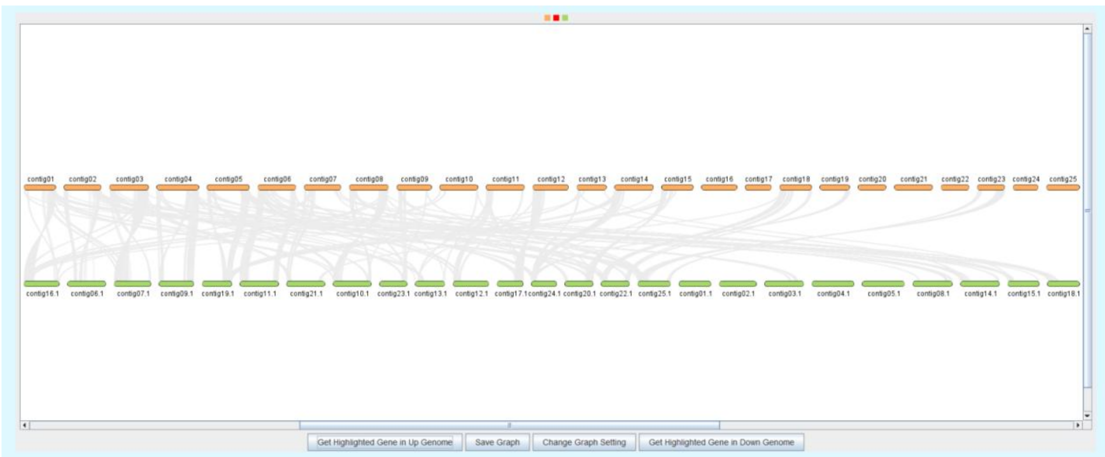


4. Next, check and confirm the chromosome order of the two genomes. Mitochondrial sequences, chloroplast sequences, fragments, and other sequences can be removed here, and chromosomes can be arranged in a user-defined order.

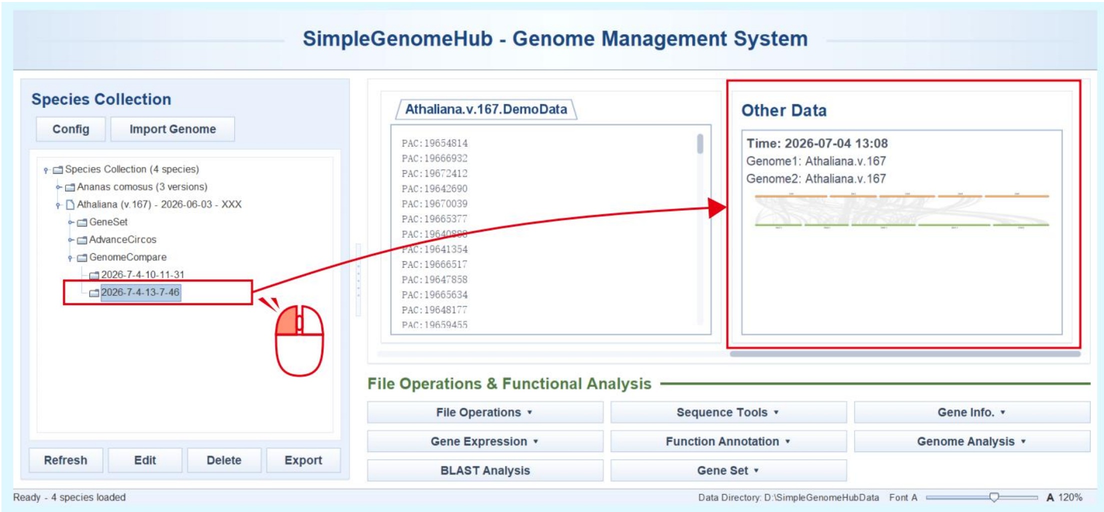
5. If there are genes that need to be highlighted, enter them under **Gene Highlight**, or select an existing gene set. After the analysis is completed, Simple Genome Hub will check whether any syntenic blocks contain the gene IDs to be highlighted.



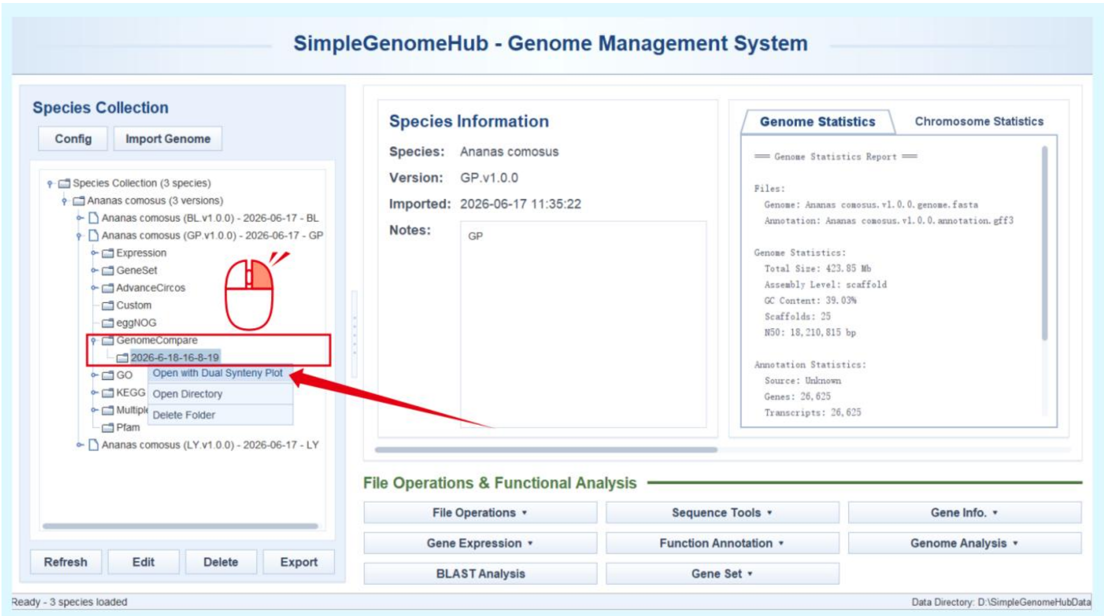
6. Click the **Start** button to run the analysis. Simple Genome Hub will invoke MCscanX (Pure Java) from TBtools-II for computation and use Dual Synteny Plot for MCScanX from TBtools-II for visualization.



- The analysis results will be automatically saved under the **GenomeCompare** directory of the selected data unit, with the folder named according to the date. And the results can be previewed on the **Other Data** interface. If the results are not displayed, click **Refresh** on the main interface.



- In the main interface, right-click the result and select **Open with Dual Synteny Plot** to visualize the saved results again using Dual Synteny Plot for MCScanX.



In practical analyses, pairwise genome synteny is the basis for other functions such as **Circos visualization** and **multi-genome synteny analysis**. Therefore, Simple Genome Hub provides interfaces for synteny results, ensuring that all synteny results generated within the database can be retrieved and rapidly invoked by other functions.

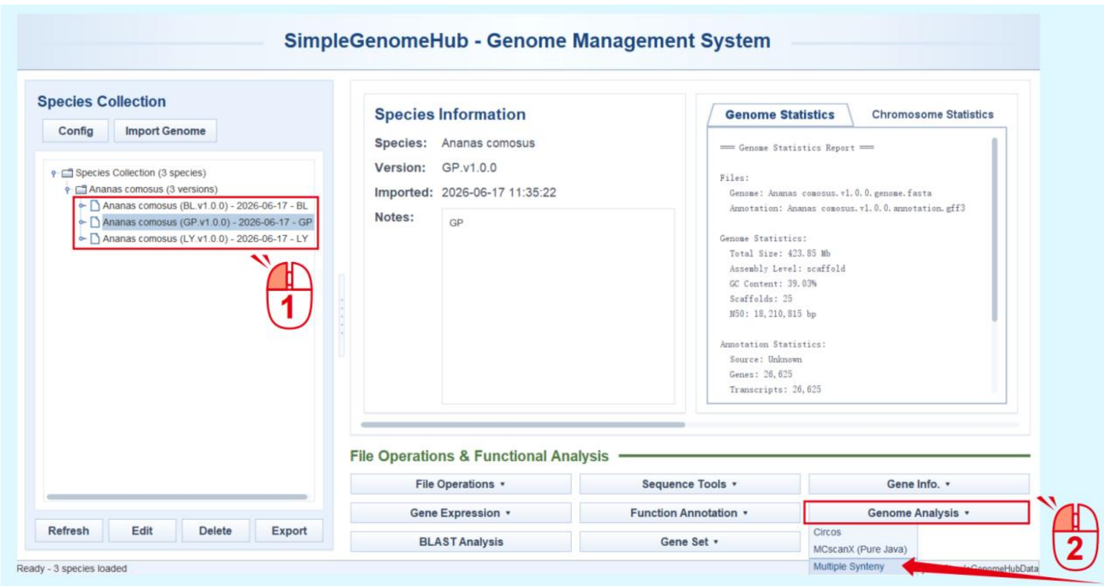
Simple Genome Hub | 10. Multi-Genome Synteny Analysis

The traditional workflow for multi-genome synteny analysis requires users to repeatedly perform pairwise synteny analyses in TBtools-II, collect Link information, manually create highlighted Links, and then generate plots through custom scripts. Moreover, users can only evaluate the final layout after all analyses have been completed. If the layout is unsatisfactory, the entire workflow often needs to be repeated multiple times, making the process highly cumbersome.

To address this issue, Simple Genome Hub introduces a Layout Workspace, which allows users to customize the layout of multiple genomes and their synteny relationships in a draft-like manner before running the analysis. This workspace supports flexible multi-chromosome layout, position snapping, Link type switching, adjustment of displayed chromosomes and their order, and other interactive operations. All layout information is recorded before execution and fully reproduced after the analysis is completed.

1. Select one of the genome data units for synteny analysis. The genomes to be compared can still be changed later, but the related results will be saved under the genome data unit selected at this step.

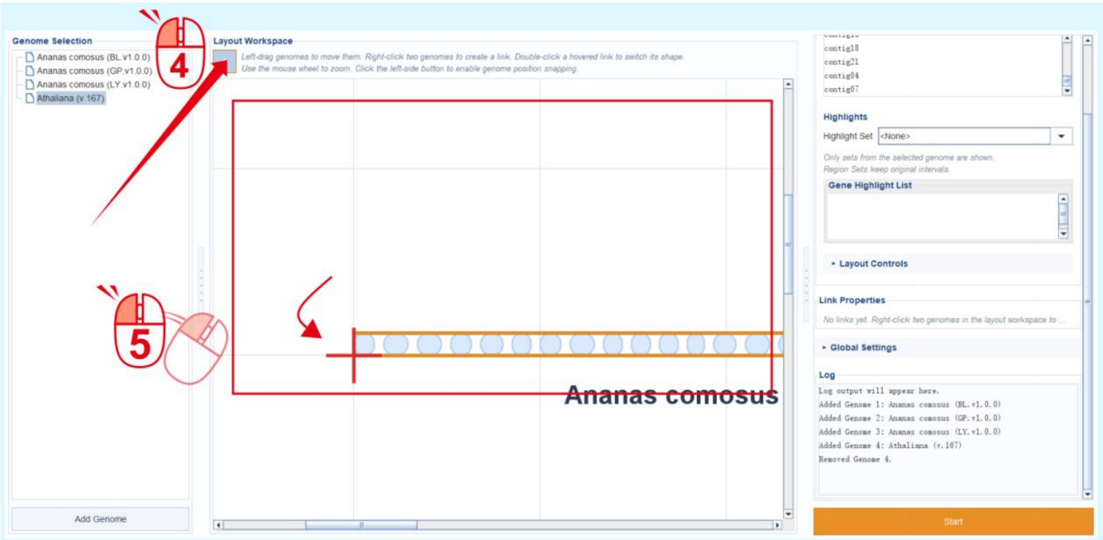
2. Click **Multiple Synteny** under **Genome Analysis**.



3. In the pop-up interface, drag the other genome data units to be analyzed into the **Layout Workspace**. Alternatively, click the **Add Genome** button to import genomes. To remove a genome, select it in the Layout Workspace and press the **Delete** key.

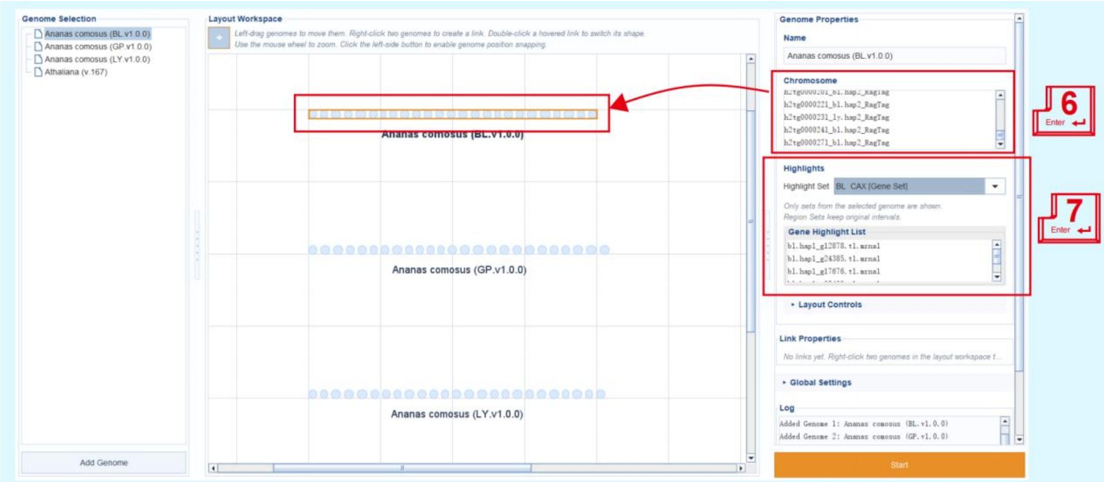
4. Click the "+" button above the Layout Workspace to enable automatic position snapping. The snapping target is the background grid with an interval of 100 units, which helps users arrange genomes at fixed spacing.

5. After position snapping is enabled, drag the genomes in the Layout Workspace to the desired positions.

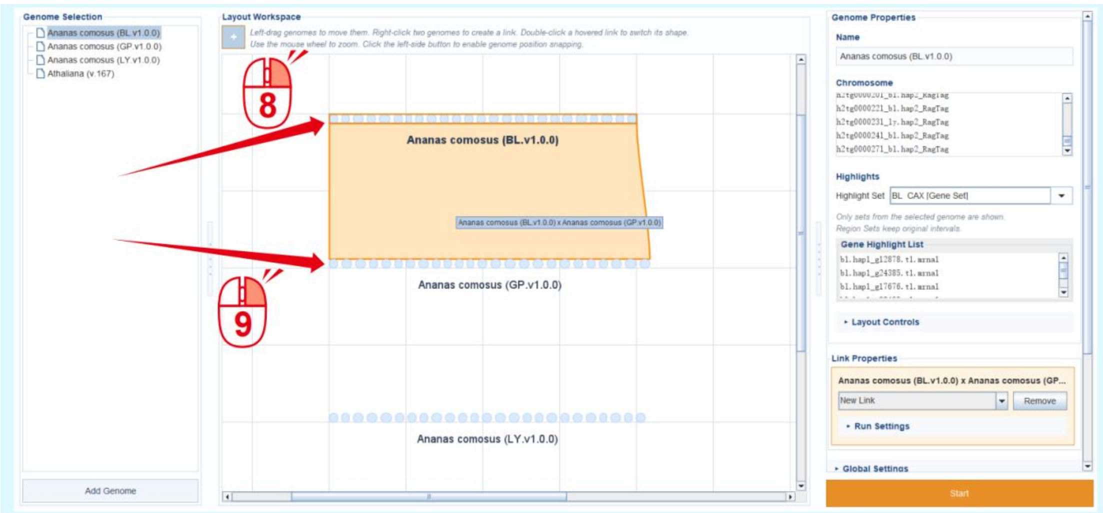


6. Click each genome in the **Layout Workspace**. In the **Genome properties** panel on the right, edit the **chromosomes** to be displayed and their order. All edits are updated in real time in the **Layout Workspace**.

7. If needed, enter the gene IDs to be highlighted on the genome, or select an existing gene set from the drop-down menu to fill them in directly.



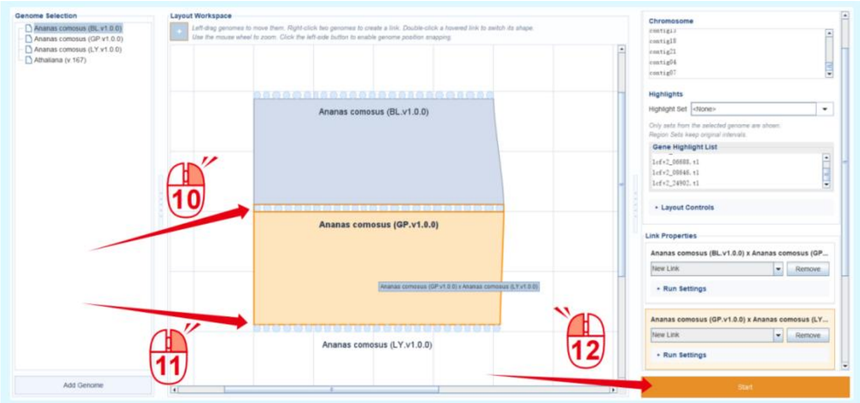
8-9. In the **Layout Workspace**, right-click the two genomes for which synteny analysis will be established to create a Link. To delete a Link, click the **Remove** button on the corresponding Link card in the Link properties panel on the right.

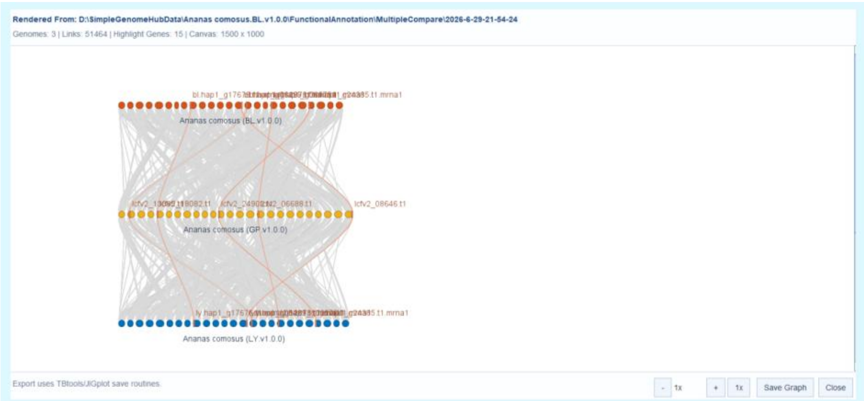


During execution, the software reads the Link information in the **Link Properties** panel sequentially. If no existing Link is selected from the drop-down menu, the software treats it as a New Link by default. The New Link workflow invokes the MCscanX (Pure Java) function of TBtools-II, automatically constructs input files according to the displayed chromosomes and chromosome order specified in the Genome properties panel, and then runs the analysis.

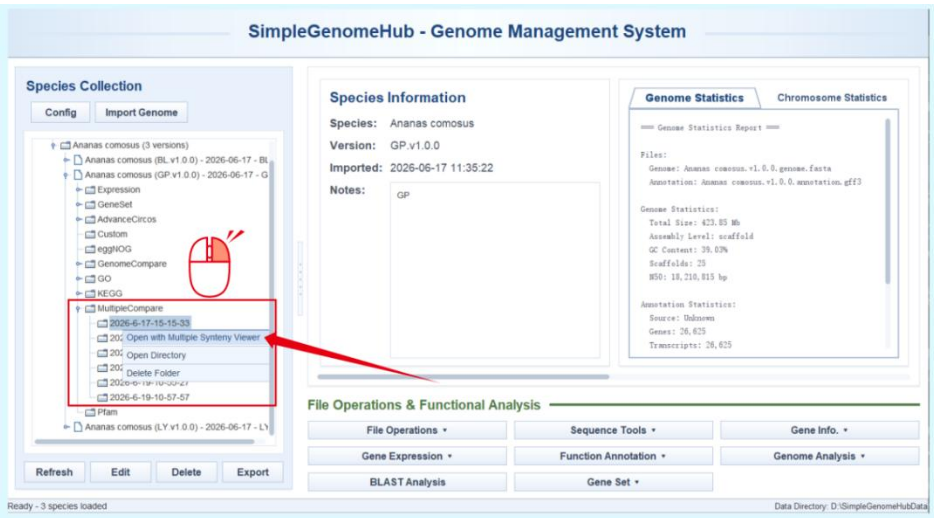
10-11. Use the same method to create a Link between GP and LY.

12. Click **Start** to begin the analysis. After the analysis is completed, the software integrates the results and constructs highlighted Links. The final visualization is then generated according to the overall layout parameters.





Each analysis result is automatically saved under the **MultipleCompare** directory of the current genome data unit, with the folder named according to the date. In the main interface, right-click the result file and select **Open with Multiple Synteny Viewer** to reopen and visualize the saved result.



Simple Genome Hub | 11. Fine-Tuning Multi-Genome Synteny Layouts

To facilitate multi-genome comparison, Simple Genome Hub introduces a Layout Workspace. For its basic usage and advantages, please refer to:

Simple Genome Hub | 10. Multi-Genome Synteny Analysis

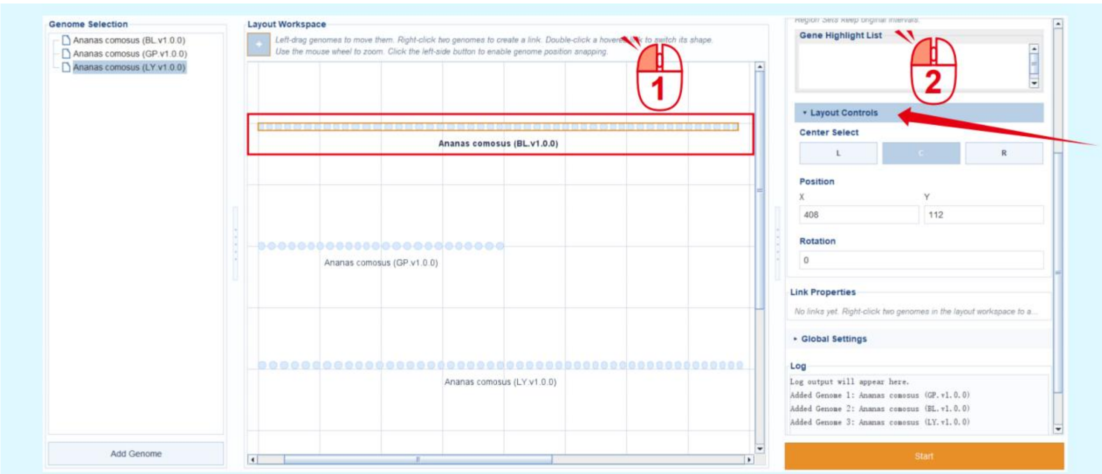
This section uses two examples to demonstrate the location and usage of layout-related parameters, helping users perform fine-scale layout adjustment for multi-genome synteny visualization.

I. Creating Syntenic Links after Genome Rotation

This example continues from **Step 7.** in **Simple Genome Hub | 10. Multi-Genome Synteny Analysis**.

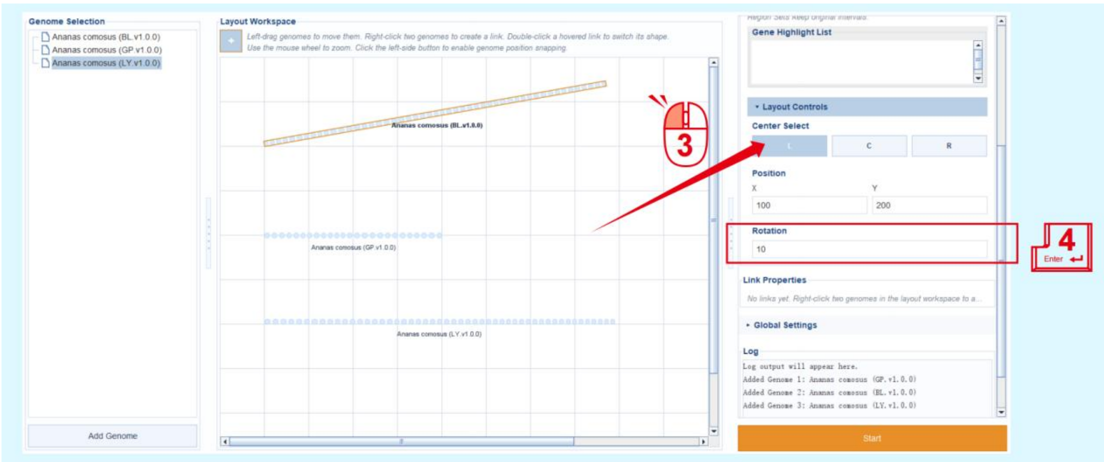
1. Select a genome track in the **Layout Workspace**.

2. Click **Layout Controls** under **Genome Properties** to expand the layout and position information of the BL genome.

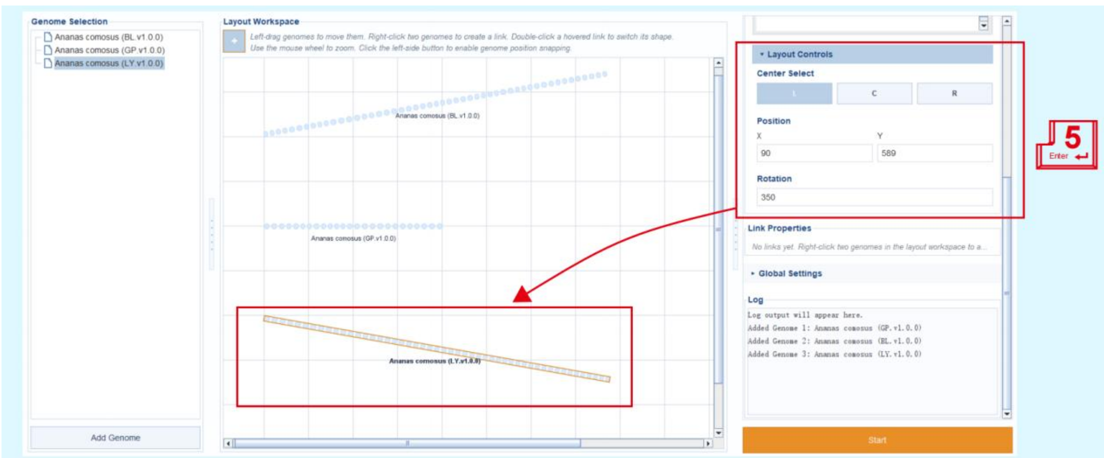


3. Select the lower-left vertex of the genome track as the reference point (Here, **L** represents the lower-left vertex of the genome track in the Layout Workspace, **C** represents the center point of the genome track, and **R** represents the lower-right vertex of the genome track).

4. Enter the rotation angle under **Rotation**. The genome track in the **Layout Workspace** will be updated in real time. In this example, the value is set to 10, meaning that **BL** is rotated 10° counterclockwise using its lower-left vertex as the reference point. The text box only recognizes numbers between 1 and 360 and does not recognize symbols such as "-". Values greater than 360 are automatically divided by 360, and the remainder is used.



5. Similarly, set LY to rotate 350° counterclockwise, which is equivalent to a 10° clockwise rotation.

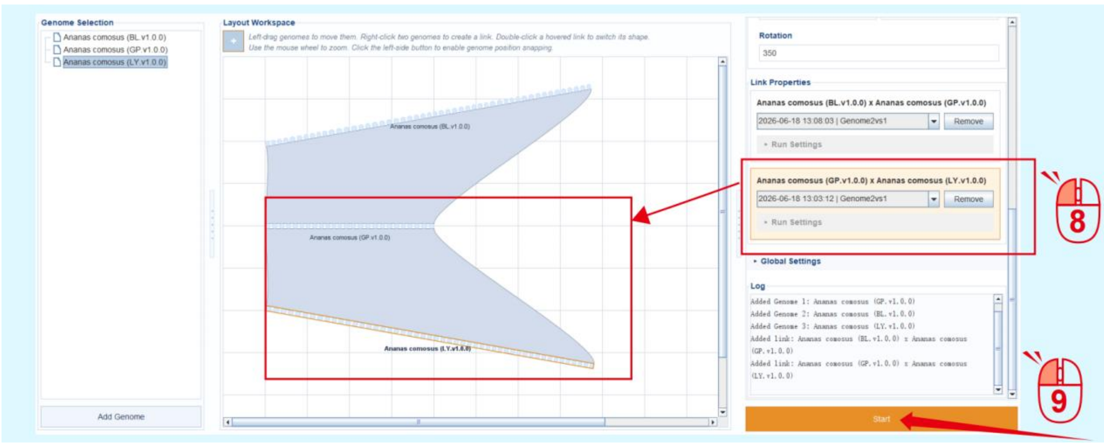


6. Create a Link between BL and GP. The Link will automatically adapt to the rotation angle of the genome tracks.

7. In **Link Properties**, because synteny analysis has already been performed previously for the same displayed chromosomes, the existing result can be selected directly, avoiding repeated computation. The displayed chromosomes only need to be consistent; their display order does not affect recognition.

8. Similarly, create the Link information between LY and GP.

9. Click **Start** to run the analysis.

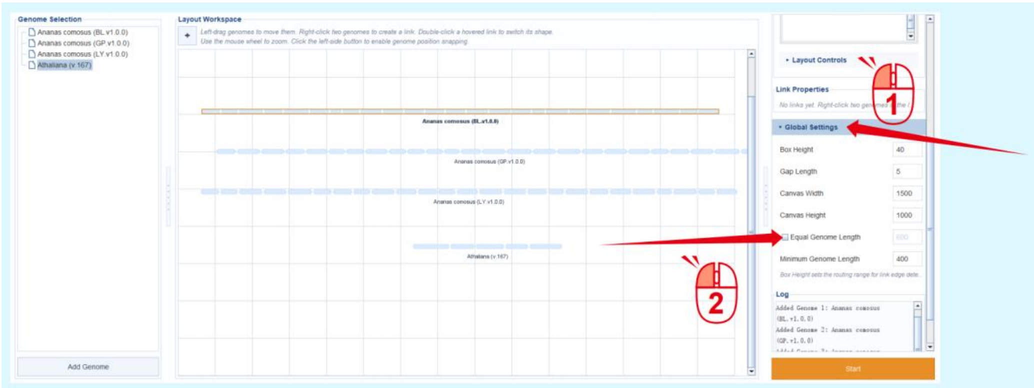


II. Advanced Genome Layout Adjustment

When the genomes to be compared differ greatly in size, such as pineapple versus *Arabidopsis thaliana*, plotting them according to their actual lengths may produce an unsatisfactory layout. In such cases, an equal-length layout can be used.

1. Click **Global Settings** to expand the global settings panel.

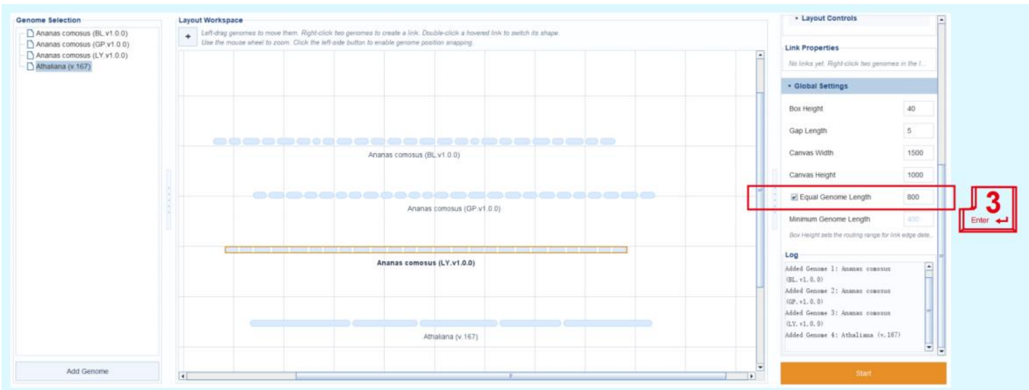
2. Check **Equal Genome Length** to make all genome tracks in the Layout Workspace have the same length.

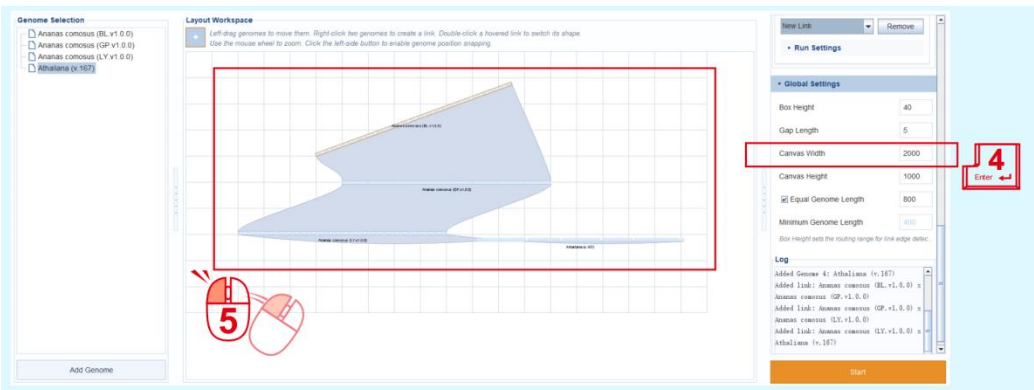


3. Set the length of all genome tracks to 800.

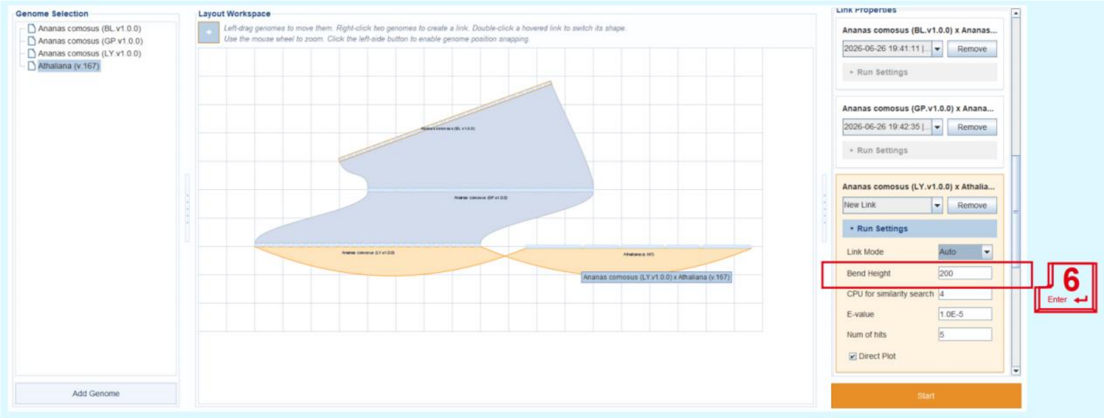
4. If the genome tracks are too large and extend beyond the canvas, adjust the canvas size. In this example, the canvas width is set to 2000.

5. Then, freely adjust the layout of the genome tracks and create Links.

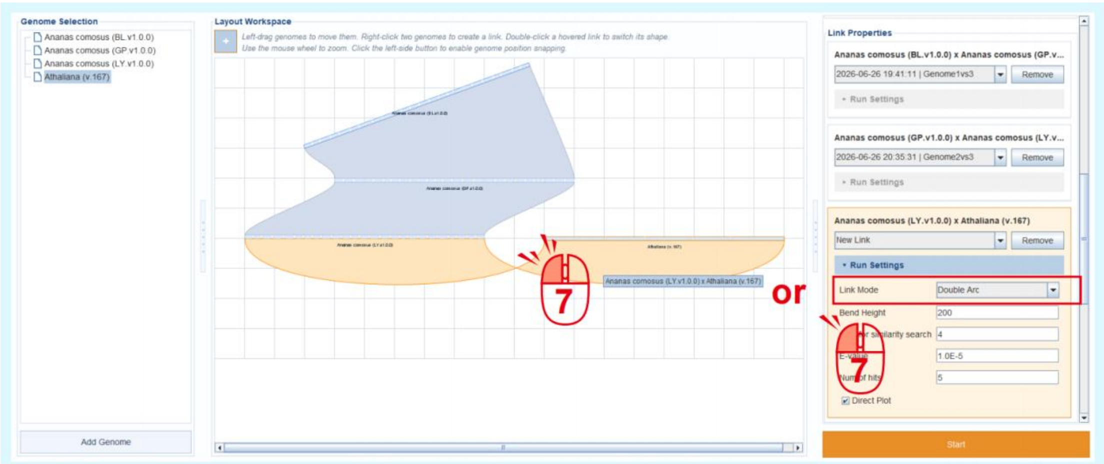




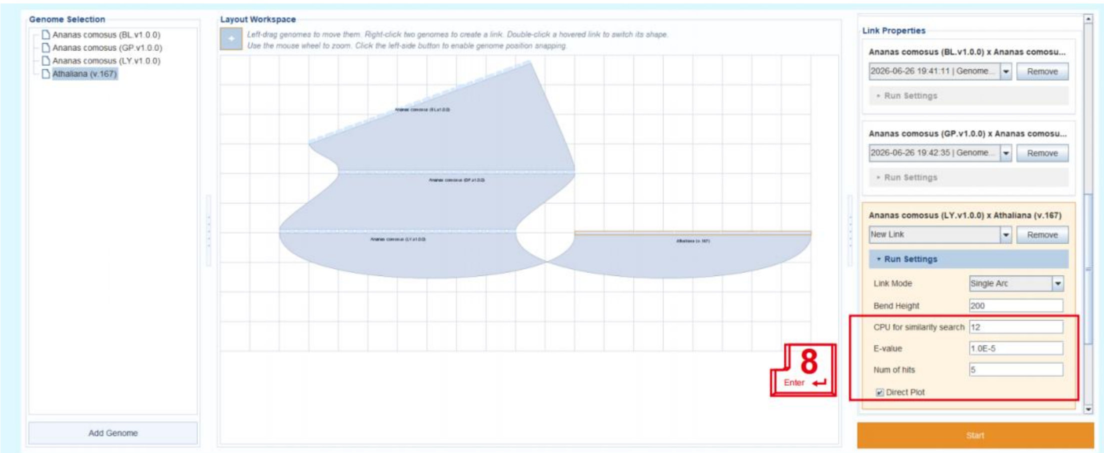
6. Adjust the curvature height of an individual Link. In this example, the value is set to 200, and the Link becomes thicker accordingly.



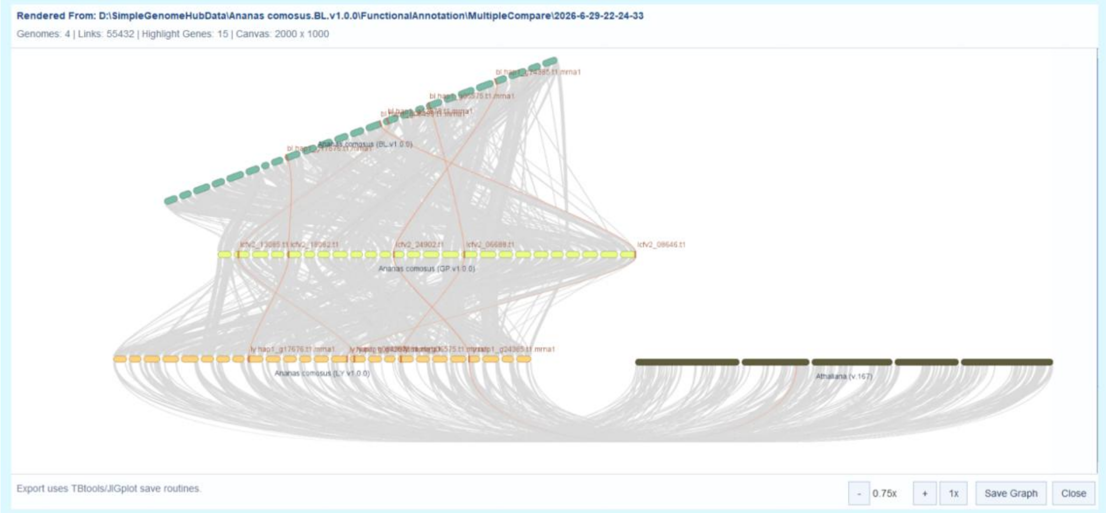
7. Double-click a Link with the left mouse button to change its type. Three Link types are available: **Auto**, which automatically determines the Link shape using collision boxes; **S-shaped**, which bends twice; and **C-shaped**, which bends once. The Link type can also be modified in **Link Properties** on the right.



8. When New Link is selected, the parameters for synteny analysis can be adjusted.



Final output:

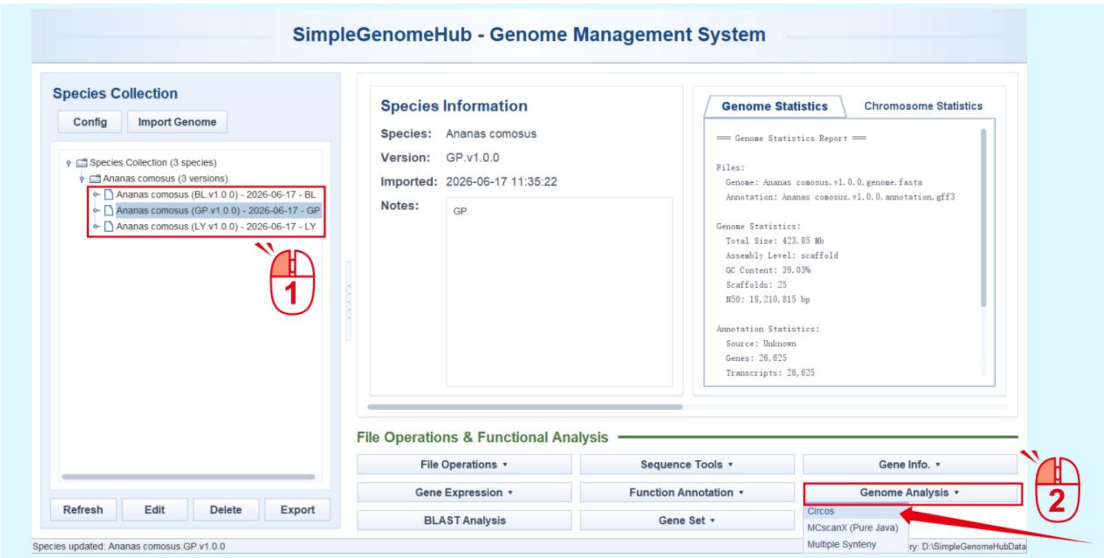


Simple Genome Hub | 12. Drawing Genome Circos Plots

Even when using TBtools-II for Circos plotting, users still need to prepare the basic genome information file, the statistical data for each track to be displayed, the Link information, and other required inputs manually. In Simple Genome Hub, however, a one-click workflow is provided for generating a basic Circos plot. The output is saved in a text format fully compatible with the Circos function in TBtools-II, and TBtools-II Advance Circos is automatically invoked for visualization.

1. Select the genome for which the Circos plot will be generated.

2. Click the **Circos** function under **Genome Analysis**.



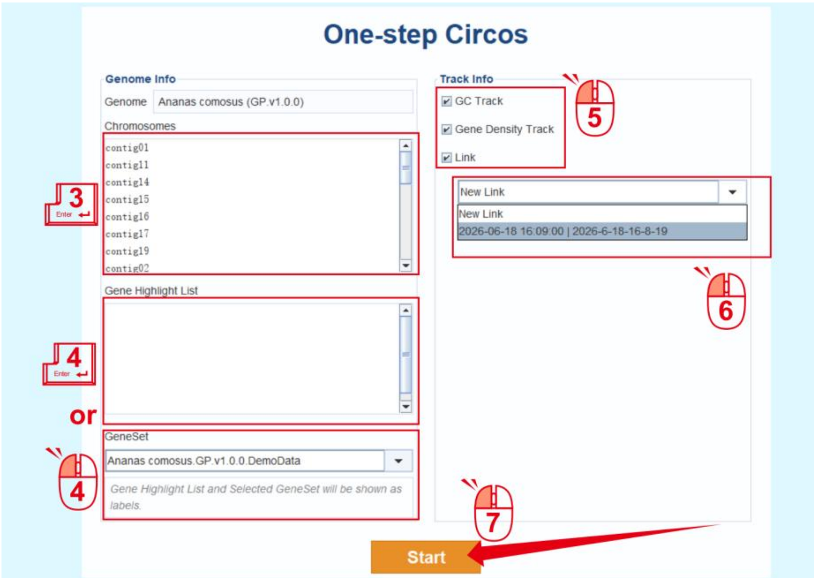
3. Edit the displayed chromosomes and their order.

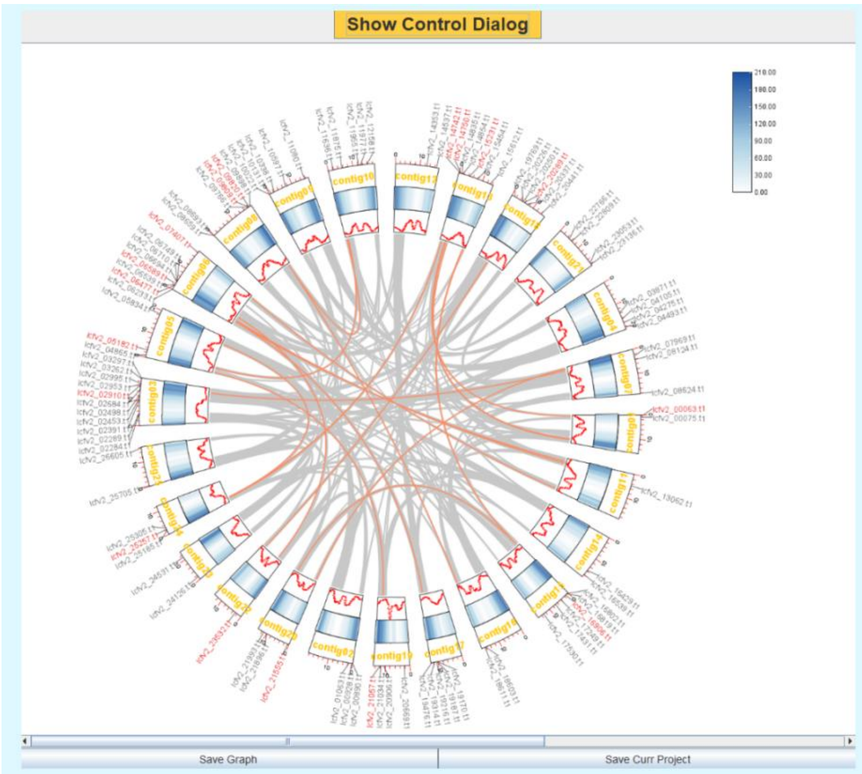
4. Enter the gene IDs to be highlighted, or select an existing gene set.

5. Select the commonly used tracks of a basic Circos plot for drawing, including **GC Track**, **Gene Density Track**, and **Link**.

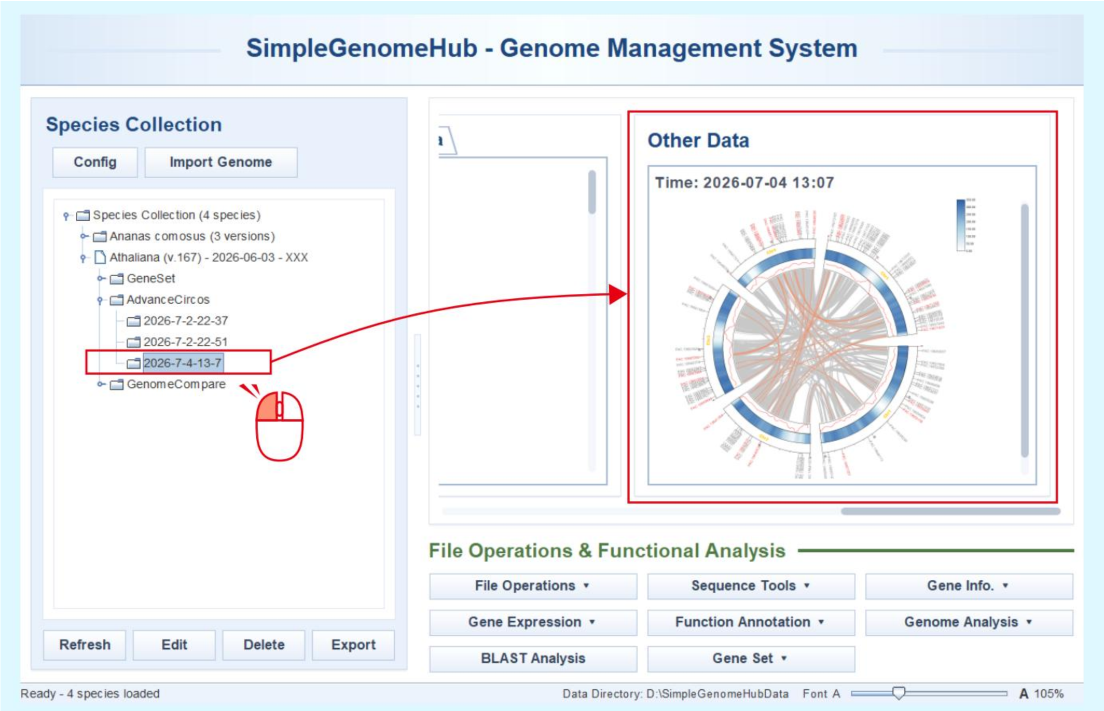
6. If a self-to-self synteny result for the selected genome already exists and matches the displayed chromosomes, it will be listed in the drop-down menu for selection. If **New Link** is selected, Simple Genome Hub will automatically invoke MCscanX (Pure Java) from TBtools-II to perform synteny analysis.

7. Click **Start** to begin the analysis.





- Each result is automatically saved under the **AdvanceCircos** directory of the current genome data unit, with the folder named according to the execution date. And the results can be previewed on the **Other Data** interface.



- In the main interface, the saved result file can be right-clicked and reopened for visualization by selecting **Open with AdvanceCircos**.